A DETERMINANT BOUND OF 498 FOR THE TENTH DISCRIMINANT

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Abstract. We address the challenge of obtaining a small explicit upper bound for a hypothetical tenth imaginary quadratic discriminant of class number one, without using Heegner's methods or the known classification. We also make no quasi-Riemann-hypothesis assumption. We prove the computer-assisted bound $D \leq 498$ for every negative fundamental discriminant $-D$ with class number one. The proof constructs a nonzero interpolation determinant from polynomial values and derivatives at algebraic integers of small absolute value. Inert primes force its norm to have a large divisor, while a spectral estimate bounds the same norm from above. A rank theorem allows the determinant to be formed using nearly the minimum number of points permitted by the dimensions of the matrix. This reduces the upper bound enough to exclude $D = 499$. The numerical comparisons are evaluated with directed rounding.

1. Introduction

Let $-D < 0$ be a fundamental discriminant, let $F_D = \mathbb{Q}(\sqrt{-D})$, and write $\mathcal{O}_D$ for its ring of integers. Gauss's class number one problem asks when $\mathcal{O}_D$ has unique factorization, or equivalently when its ideal class number $h(-D)$ is one. Heegner [6], Baker [1] and Stark [8] proved that this occurs exactly for

$$D \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}.$$

Here we prove a weaker bound by a different method.

Theorem 1.1. *Let $-D < 0$ be a fundamental discriminant with $h(-D) = 1$. Then $D \leq 498$.*

The challenge is to obtain as small an explicit upper bound as possible for a hypothetical tenth discriminant, without relying on Heegner's methods or the known classification [10]. We approach this problem through the determinant method developed in recent work on Landau–Siegel zeros [2, 9]. The remaining candidates are excluded by determinant inequalities.

1.1. **Overview of the proof.** Suppose that $h(-D) = 1$ with $D > 498$. We will form a matrix by evaluating polynomials in (z, w) and their z-derivatives at pairs $(z, \bar{z})$ with $z \in \mathcal{O}_D$. If this matrix has full row rank, it has a nonzero square minor Δ. Since its entries lie in $\mathcal{O}_D$, the norm $N = \Delta\bar{\Delta} = |\Delta|^2$ is a positive integer. We will bound N from above using the absolute values of the entries, and show that it is divisible by an integer larger than this upper bound. This contradiction will exclude the proposed value of D. The choice of polynomials and evaluation points must therefore serve both estimates while ensuring that the matrix has full row rank.

To see how the class number assumption enters, suppose that a prime p is not inert. There is then an ideal of norm p, which must be principal because $h(-D) = 1$. Its generator has norm p

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and cannot be a rational integer. But every non-rational element of $\mathcal{O}_D$ has norm at least $D/4$, so every prime $p < D/4$ must be inert. At such a prime, conjugation agrees with Frobenius modulo p, giving $z^p \equiv \bar{z} \pmod{p\mathcal{O}_D}$. Thus the polynomial $z^p - w$ takes values divisible by p at every point $(z, \bar{z})$ with $z \in \mathcal{O}_D$. By expressing the rows of our matrix in terms of powers of $z^p - w$, we obtain prime powers dividing N.

This change of basis must remain within the chosen polynomial space. We use the monomials $z^r w^s$ with $r + Ks < R$, where K and R are parameters. For $p \geq K$, replacing z^p by w does not increase $r + Ks$, so the triangular space of exponents permits all the required row operations. Taking several derivatives at each point gives more columns from the same set of lattice points. Differentiation reduces the resulting divisibility, but by an amount that can be bounded explicitly in terms of the derivative order. We also use inert primes above $D/4$: the same norm argument bounds the number of non-inert primes in an interval by the number of elements of $\mathcal{O}_D$ whose norms lie there.

For the upper bound, we choose the n lattice points of $\mathcal{O}_D$ nearest to the origin. Reducing n reduces the radius of the disk containing them and hence the bounds on the polynomial values. The estimates must also account for the derivative columns at each point. These columns are far from orthogonal, so bounding each separately loses too much at the smaller values of D. Instead, we estimate their Gram matrix, retaining the saving from these correlations before combining the contributions of the different points.

It remains to determine how few points we can use without losing full row rank. If the polynomial space has dimension M and we take J derivative orders at each point, there are nJ columns, so necessarily $nJ \geq M$. We prove a sufficient condition close to this necessary one. Direct reduction modulo an auxiliary inert prime p loses information because $z^p - w$ vanishes at every reduced node. We therefore rescale both the polynomial rows and the derivative columns using their known divisibility, obtaining an integral matrix whose reduction retains this information. A hypothetical row relation then gives a polynomial in two variables with prescribed local vanishing. Applying a differential operator and taking a determinant of coefficients produces a nonzero polynomial in one variable whose zeros, counted with multiplicity, exceed its degree. This contradiction permits a smaller n and therefore improves the upper bound for N.

We apply these estimates as follows. Elementary norm arguments first reduce $D > 498$ to primes congruent to 3 modulo 8. A uniform estimate and 101 adjacent interval comparisons exclude all $D \geq 875$, leaving sixteen possible values below 875. Fifteen are excluded by fixed determinant comparisons. The last, $D = 499$, is where the rank theorem matters: with the polynomial space and derivative orders fixed, reducing the node count from 41 793 600 to 35 394 400 raises the normalized difference between the arithmetic and size bounds from about -0.123 to about $+0.051$ (Table 4).

Section 2 introduces the determinant and the inequality we must contradict. Sections 3 and 4 develop its divisibility and size estimates. Section 5 proves the rank criterion, and Section 6 completes the numerical argument. Appendices A–C give the remaining arithmetic and spectral proofs. Appendices D–F establish the estimates used for $D \geq 875$. The code for the finite calculations is available in the accompanying repository.

2. The determinant comparison

2.1. **A model: an integer that is small and highly divisible.** Every nonzero integer N satisfies

$$\log |N| = \sum_p v_p(N) \log p. \tag{2.1}$$

Our integer will be $N = |\Delta|^2$ for a nonzero algebraic integer $\Delta \in \mathcal{O}_D$, which is the norm of Δ and hence a positive rational integer. The strategy is to bound $\log N$ from above by Archimedean

estimates, to bound $v_p(N)$ from below at many primes, and to show that the two bounds are incompatible when $h(-D) = 1$. This subsection illustrates the comparison using elementary estimates. The rest of the paper sharpens each of its three ingredients.

The matrix. We work with polynomials in two independent variables z and w. The variable w will be specialized to the complex conjugate of the value of z, but it is treated as independent of z throughout. For $P \in \mathbb{Z}[z, w]$ define the Hasse derivatives in z, with w held fixed, by

$$P(z+t,w) = \sum_{j\ge 0} (\partial_z^{[j]} P)(z,w)\, t^j, \qquad \partial_z^{[j]}(z^r w^s) = \binom{r}{j} z^{r-j} w^s.$$

Hasse derivatives keep integer coefficients; ordinary derivatives would introduce a factor $j!$. Fix a positive integer J. Let $z_1, \dots, z_n \in \mathcal{O}_D$ be distinct *nodes*, and let $\Lambda \subset \mathbb{Z}^2_{\ge 0}$ be a finite set of exponents. The *jet matrix* has rows indexed by $(r, s) \in \Lambda$, columns indexed by pairs (i, j) with $1 \le i \le n$ and $0 \le j < J$, and entries

$$V_{(r,s),(i,j)} = (\partial_z^{[j]} z^r w^s)(z_i, \overline{z_i}) = \binom{r}{j} z_i^{\,r-j}\, \overline{z_i}^{\,s} \in \mathcal{O}_D. \tag{2.2}$$

For a set C of $|\Lambda|$ columns, write Δ_C for the corresponding maximal minor. More generally, the row of any polynomial P in the span of the monomials $z^r w^s$, $(r, s) \in \Lambda$, is the vector of its jets $(\partial_z^{[j]} P)(z_i, \overline{z_i})$. Integral unitriangular changes of the spanning polynomials leave every Δ_C unchanged.

One Frobenius calculation. The divisibility comes from the behaviour of conjugation modulo an inert prime.

Lemma 2.1. *Let p be a prime that is inert in F_D, and let $z \in \mathcal{O}_D$. Then $z^p \equiv \overline{z} \pmod{p\mathcal{O}_D}$. Consequently, if $g = z^p - w$, then*

$$g(z + t, \overline{z}) = pU(t) + t^p$$

for a polynomial $U \in \mathcal{O}_D[t]$ of degree less than p, and for all integers $m, j \ge 0$,

$$p^{\lfloor j/p \rfloor}\, [t^j]\, g(z + t, \overline{z})^m \in p^m \mathcal{O}_D. \tag{2.3}$$

Proof. The quotient $\mathcal{O}_D/p\mathcal{O}_D$ is the field of p^2 elements. Conjugation induces its nontrivial automorphism, which is the pth-power map. This gives $z^p \equiv \bar{z} \pmod{p\mathcal{O}_D}$, including when $p = 2$. Next,

$$(z+t)^p - \overline{z} = (z^p - \overline{z}) + \sum_{k=1}^{p-1} \binom{p}{k} z^{p-k} t^k + t^p,$$

and every term except t^p is divisible by p. Finally, $(pU + t^p)^m = \sum_{q=0}^{m} \binom{m}{q} p^{m-q} U^{m-q} t^{pq}$, and a term contributes to t^j only if $q \le \lfloor j/p \rfloor$. Each such term is divisible by $p^{m-\lfloor j/p \rfloor}$, which proves (2.3). □

Geometrically, the lemma says that, p-adically, every point $(z, \overline{z})$ lies close to the curve $z^p = w$ [9]. The polynomial g is small at every node, and its z-derivatives are small too, since $\partial_z g = p z^{p-1}$. This is why the jets differentiate only in z: a derivative in w would not preserve the divisibility.

Where the triangle comes from. To use Lemma 2.1, write each row exponent as $r = mp + a$ with $0 \le a < p$ and replace the monomial $z^r w^s$ by

$$g^m z^a w^s = z^r w^s + \sum_{i=1}^{m} (-1)^i \binom{m}{i} z^{r-pi} w^{s+i}. \tag{2.4}$$

By the Leibniz rule and (2.3), every jet of $g^m z^a w^s$ of order j, multiplied by $p^{\lfloor j/p \rfloor}$, lies in $p^m \mathcal{O}_D$. The change of basis (2.4) is unitriangular for the ordering by z-degree, but only if every exponent $(r - pi, s + i)$ on the right is again in Λ. The support must therefore be closed under the Frobenius substitution $z^p \mapsto w$.

The rectangle $\{r < R,\ s < S\}$ used by Green is not closed under this substitution: it is stopped by the ceiling $s < S$. There the substitution can be applied only $\min(\lfloor r/p \rfloor, S - 1 - s)$ times, and the p-adic credit of the row is truncated accordingly [2, Lemma 5.1]. The remedy is to grade the monomials so that the substitution never raises the degree. Fix integers $K \ge 2$ and $S \ge 1$. Give z weight 1 and w weight K, and take

$$\Lambda = \{(r, s) \in \mathbb{Z}^2_{\ge 0} :\ r + Ks < R\}, \qquad R = KS, \qquad M = |\Lambda| = \frac{R(S+1)}{2}. \tag{2.5}$$

The substitution $z^p \mapsto w$ changes the weight by $K - p$, so it never leaves Λ when $p \ge K$. Every row then receives its full credit $\lfloor r/p \rfloor$. Primes $p < K$ receive truncated credit, and that boundary is treated in Section 3.

Scaling column (i, j) by $p^{\lfloor j/p \rfloor}$ multiplies Δ_C by $p^{\sum_{(i,j)\in C} \lfloor j/p \rfloor}$. Since $p\mathcal{O}_D$ is prime and $v_p(|\Delta_C|^2) = 2 v_{p\mathcal{O}_D}(\Delta_C)$, we obtain, for every inert prime $p \ge K$,

$$v_p(|\Delta_C|^2) \ \ge\ 2 \sum_{(r,s)\in\Lambda} \Big\lfloor \frac{r}{p} \Big\rfloor \ - \ 2 \sum_{(i,j)\in C} \Big\lfloor \frac{j}{p} \Big\rfloor. \tag{2.6}$$

The row contribution, including its factor 2, is about W/p, where $W = R^2 S/3 \approx 2 \sum_\Lambda r$. The second is a column loss that is present only when derivative orders reach p. For the smaller primes used below, J is much larger than p, so this loss cannot be neglected. Section 3 shows how it combines with the factorial sizes of the Hasse coefficients.

Sizes. If every node satisfies $|z_i| \le \rho$ with $\rho \ge 1$, then each entry of row (r, s) has absolute value at most $\binom{r}{j} \rho^{r+s-j}$. Hadamard's inequality therefore gives

$$\log |\Delta_C|^2 \ \le \sum_{(r,s)\in\Lambda} 2(r+s) \log \rho \ + \sum_{(r,s)\in\Lambda} \log \sum_{(i,j)\in C} \binom{r}{j}^2 \rho^{-2j}. \tag{2.7}$$

The first term is $(1 + 1/K + O(1/S))\, W \log \rho$. Comparing this with the divisibility bound (2.6) gives a necessary inequality for every nonzero minor. After division by W, its arithmetic side is roughly the sum of $(\log p)/p$ over inert primes $K \le p < R$, less the derivative loss; its size bound is roughly $(1 + 1/K) \log \rho$, plus the derivative term. The class number assumption bounds the missing prime contributions through the norm form, as we explain in Section 3.

This model already shows where the difficulty lies, and it points to the three refinements. First, the arithmetic credit must be computed exactly, including small primes, prime powers, and the loss from non-inert primes, bounded through the norm form (Section 3). Second, Hadamard's inequality treats the entries of each node block as unrelated and discards their cancellation. The spectral estimates of Section 4 recover part of it. Because the nonzero minor is not identified, those estimates must hold for every column set C. Third, ρ is governed by the node count. The n nearest lattice points lie in a disk of radius about $\sqrt{n\sqrt{D}/(2\pi)}$ (Lemma 2.5), and $n \ge M/J$ is forced.

2.2. **Arithmetic of class number one.** Every element of $\mathcal{O}_D$ has the form $\alpha = (x + y\sqrt{-D})/2$ with $x, y \in \mathbb{Z}$ and $x \equiv yD \pmod 2$, and its norm is

$$N(\alpha) = |\alpha|^2 = \frac{x^2 + Dy^2}{4}. \tag{2.8}$$

The elementary reductions below follow from one observation about norms.

Lemma 2.2 (Norm obstruction)**.** *If $\alpha \in \mathcal{O}_D \setminus \mathbb{Z}$, then $N(\alpha) \geq D/4$. If $\alpha \in \mathbb{Z}$, then $N(\alpha)$ is a square.*

Proof. If $\alpha \notin \mathbb{Z}$ then $y \neq 0$ in (2.8), so $N(\alpha) \geq Dy^2/4 \geq D/4$. If $\alpha \in \mathbb{Z}$ then $N(\alpha) = \alpha^2$. □

Lemma 2.3. *Suppose $h(-D) = 1$. Then every prime $p < D/4$ is inert in F_D.*

Proof. If p is split or ramified, there is a prime ideal of norm p. It is principal, generated by some α with $N(\alpha) = p$. Since p is not a square, $\alpha \notin \mathbb{Z}$, and Lemma 2.2 gives $p \geq D/4$. □

Lemma 2.4 (Reduction)**.** *Suppose $h(-D) = 1$ and $D > 16$. Then D is a prime and $D \equiv 3 \pmod 8$.*

Proof. The ramified primes are the prime divisors of D, so by Lemma 2.3 every prime divisor of D is at least $D/4$. If D is even, then 2 ramifies and $2 < D/4$, which is impossible. Hence D is odd, and since $-D$ is a fundamental discriminant, D is squarefree with $D \equiv 3 \pmod 4$. If D were composite, its least prime factor q would satisfy $q \leq \sqrt{D} < D/4$, again impossible. So D is prime. Finally, if $D \equiv 7 \pmod 8$ then $-D \equiv 1 \pmod 8$ and 2 splits, contradicting Lemma 2.3. Therefore $D \equiv 3 \pmod 8$. □

For the proof of Theorem 1.1 we assume $D > 498$ from now on, so D is a prime congruent to 3 modulo 8, and both 2 and every odd prime below $D/4$ are inert. Since $D \equiv 3 \pmod 4$, the integer cutoff $p < D/4$ is equivalently $p \leq (D-3)/4$. We write $\omega = (1 + \sqrt{-D})/2$, so that $\mathcal{O}_D = \mathbb{Z}[\omega]$. In these coordinates $\mathrm{N}(x_0 + y\omega) = (x_0 + y/2)^2 + Dy^2/4$; it is the same norm form as (2.8), with $x = 2x_0 + y$.

2.3. **Nearest nodes.** Our nodes are the lattice points of $\mathcal{O}_D \subset \mathbb{C}$ nearest to the origin. Let $z_1, z_2, \ldots$ be the elements of $\mathcal{O}_D$ listed with $|z_1| \leq |z_2| \leq \cdots$, with ties broken arbitrarily. By (2.8), $\mathcal{O}_D$ is the union of the horizontal rows

$$\left\{ \tfrac{x}{2} + \tfrac{y}{2}\sqrt{-D} : \ x \equiv yD \pmod 2 \right\}, \qquad y \in \mathbb{Z}.$$

The rows are spaced $\sqrt{D}/2$ apart vertically, and within each row the points form a translate of $\mathbb{Z}$. The covolume of $\mathcal{O}_D$ is therefore $\sqrt{D}/2$.

Lemma 2.5 (Nearest-node disk)**.** *Put $\delta = \sqrt{D+4}/4$. For every $i \geq 1$,*

$$|z_i| \ \leq \ \rho_i := \sqrt{\frac{i\sqrt{D}}{2\pi}} + \delta. \tag{2.9}$$

Proof. Every point of the plane is within vertical distance $\sqrt{D}/4$ of a lattice row, and within horizontal distance $1/2$ of a point on that row. Thus the lattice has covering radius at most δ. For $z \in \mathcal{O}_D$, let $V(z)$ be its Voronoi cell, the set of points of $\mathbb{C}$ at least as close to z as to any other lattice point. These cells are translates of one another, they cover $\mathbb{C}$, and they overlap only on their boundaries. Each has area $\sqrt{D}/2$. Every point of $V(z)$ lies within δ of z.

Let $\rho > \delta$. If $|\zeta| \le \rho - \delta$ and $\zeta \in V(z)$, then $|z| \le |\zeta| + \delta \le \rho$. Hence the closed disk of radius $\rho - \delta$ is covered by the cells of lattice points of modulus at most ρ, and comparing areas gives

$$\#\{z \in \mathcal{O}_D : \ |z| \le \rho\} \ \ge \ \frac{2\pi(\rho-\delta)^2}{\sqrt{D}}.$$

With $\rho = \rho_i$ the right side equals i, so at least i lattice points have modulus at most ρ_i. This gives $|z_i| \le \rho_i$. □

We use Lemma 2.5 in two ways. It supplies the radius $\rho = \rho_n$ in the Archimedean estimates. It also controls residues, through the following consequence.

Corollary 2.6 (Distinct residues)**.** *Let p be a prime with $2\rho_n < p$. Then $z_1, \dots, z_n$ have distinct residues modulo $p\mathcal{O}_D$.*

Proof. A nonzero element $p\beta$ of $p\mathcal{O}_D$ has modulus $p\sqrt{N(\beta)} \ge p$. The difference of two distinct nodes is nonzero and has modulus at most $2\rho_n < p$. □

2.4. **The inequality to be proved.** We now give the comparison in a form that separates the algebra from the numerical estimates. The notation introduced here will be used throughout the paper. Write

$$W = \frac{R^2 S}{3}, \qquad W_1 = 2\sum_{(r,s)\in\Lambda} r, \qquad W_2 = 2\sum_{(r,s)\in\Lambda} s. \tag{2.10}$$

The factor W is a convenient normalization: the main logarithmic terms in both estimates have size $W \log D$. Direct summation gives

$$\frac{W_1 + W_2}{W} = 1 + \frac{1}{K} + \frac{3(K-1)}{2KS} + \frac{K-5}{2KS^2}. \tag{2.11}$$

Choose M columns C for which the square submatrix V_C is nonsingular, and put $\Delta_C = \det V_C$. The rank argument will supply such a choice; it will not specify the columns. The arithmetic estimates therefore have to hold for every possible choice C. They take the form

$$W\mathcal{A} - 2\sum_{(i,j)\in C} b(j) \ \le \ \log|\Delta_C|^2. \tag{2.12}$$

Here $\mathcal{A}$ is a column-independent lower bound assembled from local divisibilities, and $b(j)$ is the loss from using a derivative of order j. Both depend on the chosen parameters. Section 3 describes their construction and states the local estimate used below.

The main part of $b(j)$ is the first-prime-power sum $\sum_q \lfloor j/q \rfloor \log q$. Legendre's formula shows that subtracting it from $\log(j!)$ leaves higher-prime-power savings, with additional contributions from non-inert primes. This remaining function is nearly linear in j; we subtract a slope and bound its finite remainder. Set

$$F(j) = \log(j!) - b(j), \qquad H(j) = j\sigma - F(j), \tag{2.13}$$

where σ is a slope chosen in the arithmetic calculation. Let c be any bound such that

$$2\sum_{(i,j)\in C} H(j) \le Mc \quad \text{for every } |C| = M. \tag{2.14}$$

There are only n columns of any one order j, so this is a finite selection problem with a capacity constraint. It need not be bounded by the much cruder $2M \max_j H(j)$.

Let ρ contain all the nodes, put $u_i = z_i/\rho$, and define

$$B_{(r,s),(i,j)}(\ell) = \frac{(r)_j}{R^j} e^{\ell j} u_i^{r-j} \bar{u}_i^{s}, \qquad \ell = \log(R/\rho) - \sigma. \tag{2.15}$$

Here $(r)_{\underline{j}} = r(r-1)\cdots(r-j+1)$ is the falling factorial, with $(r)_{\underline{0}} = 1$; the entry is zero when $j > r$. The exact relation to the integral matrix is

$$B_C(\ell) = \operatorname{diag}_\Lambda(\rho^{-r-s})\, V_C\, \operatorname{diag}_C(j!(\rho/R)^j e^{\ell j}). \tag{2.16}$$

Taking squared absolute determinants, substituting (2.13), and then using (2.12) gives

$$W\mathcal{A} \le (W_1 + W_2)\log\rho + \log|\det B_C(\ell)|^2 + 2\sum_C H(j).$$

This calculation explains the normalization: the factorials are cancelled exactly, rather than estimated twice.

Proposition 2.7 (Determinant comparison)**.** *Suppose that V has full row rank, that (2.12) and (2.14) hold, and that*

$$\frac{1}{W}\log|\det B_C(\ell)|^2 \le U_{\mathrm{var}} \qquad (|C| = M)$$

holds uniformly over all column selections. Then

$$\mathcal{A} \le \frac{W_1 + W_2}{W}\log\rho + U_{\mathrm{var}} + \frac{Mc}{W}. \tag{2.17}$$

In particular, a strictly larger arithmetic lower bound is impossible.

Proof. Choose a nonsingular minor and apply the preceding identities. All bounds are uniform in the choice, so no information about which minor is nonsingular is needed. □

For the numerical comparisons we use the slightly weaker bound obtained by adding $2M\log M/W$ to the right side of (2.17). This positive term is less than 3×10^{-7} at the parameters for 499. Thus define

$$\mathcal{U} = \frac{W_1 + W_2}{W}\log\rho + U_{\mathrm{var}} + \frac{Mc + 2M\log M}{W}, \qquad G = \mathcal{A} - \mathcal{U}. \tag{2.18}$$

A positive lower bound for G rules out the proposed field. There are three separate tasks: prove full rank, establish the local arithmetic estimates, and bound the same minor at the complex embedding. The auxiliary-prime scalings used to prove rank in Section 5 do not enter (2.16); after nonvanishing has been established, all size and divisibility estimates concern the original integral minor Δ_C.

It is already possible to see why the number of nodes matters. The disk estimate gives $\rho \asymp n^{1/2}D^{1/4}$ once its boundary error is small. Reducing n therefore lowers the common term in (2.18) by approximately

$$\frac{1}{2}\left(1 + \frac{1}{K}\right)\log\frac{n_{\mathrm{old}}}{n_{\mathrm{new}}}.$$

At the two node counts in Table 4, this is about 0.085. The remaining terms also change: there are fewer copies of each derivative order, and the proportion $\theta = M/(nJ)$ of available columns that must be selected increases. We call this proportion the *occupancy*. Section 4 exploits these restrictions. In the actual calculation the combined gain is enough to change the sign of G.

3. Divisibility of the minor

We write q for a general prime, reserving p for the auxiliary prime in Section 5. The basic Frobenius argument in Section 2.1 gives a row contribution from each inert prime and a loss depending on the derivative orders selected. We now strengthen the row contribution, first by using the conjugate variable and then by retaining the small primes. Finally we explain how class number one controls the losses from non-inert primes.

Recall that for an inert prime $q \geq K$, the integral row basis is $P_{r,s} = z^{r \bmod q}(z^q - w)^{\lfloor r/q \rfloor} w^s$. Writing v_q for the valuation normalized by $v_q(q) = 1$, the calculation in Lemma 2.1 gives

$$v_q(\partial_z^{[j]} P_{r,s}(z_i, \bar{z}_i)) \geq \max(0, \lfloor r/q \rfloor - \lfloor j/q \rfloor). \tag{3.1}$$

Each determinant term uses every row and every selected column once. Hence

$$v_q(\Delta_C) \geq \sum_{\Lambda} \lfloor r/q \rfloor - \sum_{C} \lfloor j/q \rfloor. \tag{3.2}$$

Its usefulness emerges after summing over primes: the column loss combines with the factorials in the Hasse derivatives, as in (2.13).

3.1. **Additional divisibility in the conjugate variable.** Differentiation in z leaves polynomials in w untouched. Such polynomials can contribute additional divisibility at no derivative cost. We use the following elementary tower. Define

$$T_1(w) = w^{q^2} - w, \qquad \nu_a = \frac{q^{2a} - 1}{q^2 - 1}, \qquad T_{a+1}(w) = T_a(w)^{q^2} - q^{\nu_a(q^2-1)} T_a(w).$$

These are monic polynomials of degrees q^{2a}. Inductively,

$$T_a(\bar{z}_i) \in q^{\nu_a} \mathcal{O}_D. \tag{3.3}$$

Indeed, if $T_a(\bar{z}_i) = q^{\nu_a} b$, then the next value equals $q^{q^2 \nu_a}(b^{q^2} - b)$, which is divisible by $q^{q^2 \nu_a + 1} = q^{\nu_{a+1}}$.

Write $s = \sum_{a \geq 0} d_a q^{2a}$ in base q^2, and put

$$A_s(w) = w^{d_0} \prod_{a \geq 1} T_a(w)^{d_a}.$$

It is monic of degree s, and (3.3) gives

$$v_q(A_s(\bar{z}_i)) \geq \sum_{a \geq 1} d_a \nu_a = \sum_{a \geq 1} \left\lfloor s/q^{2a} \right\rfloor.$$

Replacing w^s by $A_s(w)$ in the previous basis preserves triangularity. We obtain the stronger row estimate

$$v_q(\partial_z^{[j]}(z^{r \bmod q}(z^q - w)^{\lfloor r/q \rfloor} A_s(w))(z_i, \bar{z}_i)) \geq \max(0, \lfloor r/q \rfloor - \lfloor j/q \rfloor) + \sum_{a \geq 1} \left\lfloor s/q^{2a} \right\rfloor. \tag{3.4}$$

The row contributions are finite sums over the triangle. Their exact floor-sum formulas are given in Appendix A.2.

3.2. **Small primes and non-inert primes.** For an inert prime $q < K$, replacing z^q by w increases weighted degree. The number of replacements in a row is therefore bounded by

$$h_q(r, s) = \min\left(\lfloor r/q \rfloor, \lfloor (R - 1 - r - Ks)/(K - q) \rfloor\right). \tag{3.5}$$

There is still a substantial contribution: by (A.7), $2\sum_{\Lambda} h_q(r, s) = W_2 = R(S^2 - 1)/3$.

The small-prime estimate used in the computation improves this elementary bound by combining $z^q - w$, $w^q - z$, and $z^{q^2} - z$. The first factor can lose a power of q only after spending at least q orders of differentiation; the latter two can lose one after spending one order. Optimizing where each type is used is a finite triangular row calculation. The resulting finite lower bound is recorded in Appendix A, where the choice of the cut is explained. In particular, $q = 2$ is included. Working in the maximal order is essential here, since 2 is inert when $D \equiv 3 \pmod 8$.

The arithmetic calculation also uses primes above $(D-3)/4$, where inertness is no longer guaranteed. Class number one still supplies useful information. If q splits or ramifies, an ideal of norm q is principal, so

$$q = \mathrm{N}(x_0 + y\omega) = \left(x_0 + \frac{y}{2}\right)^2 + \frac{D}{4}y^2, \qquad y \neq 0. \tag{3.6}$$

At a fixed norm cutoff there are only finitely many such pairs. One can therefore start with an all-prime estimate and subtract an upper bound for the contribution of possible non-inert primes, using these norm representations.

Counting all norm values as possible primes overestimates the loss, since composite values are charged as well. Unique factorization identifies those values: Lemma B.1 gives the number of primitive representatives of a composite norm, and Lemma B.2 shows that the required lists of factors are complete. Removing the charges for composite values with two or three prime factors increases $\mathcal{A}$; these are the only composite corrections used in Section 6.

For orientation, the resulting arithmetic term has the form

$$\mathcal{A} = \mathcal{A}_0 - L_{\mathrm{norm}} + C_2 + C_3. \tag{3.7}$$

The first term sums inert-prime row contributions as a provisional all-prime bound; L_{norm} pays for norm values that might be non-inert primes; C_2 and C_3 remove the loss assigned to composite norm values. Split and ramified primes also have their own local conjugate-variable contributions. These, and the retained derivative terms, are included in the arithmetic lower bound.

For the comparisons in Section 6, we take $\mathcal{A}$ and $b(j)$ from Proposition B.3, the slope σ from (B.13), and the remainder bound c from (B.17). These give (2.12) and (2.14) at each chosen node count. The estimates and their hypotheses are proved in Appendix B.

4. An upper bound for every column selection

We turn to the complex absolute value of the minor. An entrywise bound followed by Hadamard's inequality is enough for a large discriminant bound. At the smaller values of D considered here, however, it loses too much information about the derivative columns at a single node. Those columns are far from orthogonal. Our estimate keeps them together, bounds their Gram matrix, and then combines the node contributions.

Full row rank gives a nonsingular minor but does not tell us how many of its columns come from each node, or which derivative orders are selected. The estimates below therefore allow an arbitrary set of orders at each node, using only that there are at most J columns per node and M in total.

4.1. Weights and node Gram matrices.

Let $B(\ell)$ be the normalized matrix in (2.15). Choose real numbers $\lambda_{r,s}$ and form

$$Y = \mathrm{diag}_{\Lambda}(e^{-\lambda_{r,s}/2})B(\ell).$$

Denote by Y_i the block of J columns at node i, and write $G_i = Y_i^* Y_i$. If C_i is the set of orders selected at that node, Fischer's inequality gives

$$\log|\det B_C(\ell)|^2 \leq \sum_{\Lambda} \lambda_{r,s} + \sum_{i=1}^{n} \log\det G_i[C_i, C_i]. \tag{4.1}$$

Indeed, $Y_C^* Y_C$ is a positive semidefinite Gram matrix, and its determinant is at most the product of the determinants of its diagonal blocks. The weights are free parameters, but their total cost $\sum \lambda_{r,s}$ must be restored. Some weights have negative logarithmic cost; this signed sum will be estimated separately.

Lemma 4.1 (Radial monotonicity)**.** *For fixed row weights and a fixed set of derivative orders, each principal minor of a node Gram matrix is a polynomial with nonnegative coefficients in $q = |u_i|^2$.*

Proof. Write $u_i = \sqrt{q}\, e^{\mathrm{i}\phi}$. The phase of an entry of Y_i is $e^{\mathrm{i}(r-s-j)\phi}$, so it is removed by diagonal unitary transformations on the rows and columns. By Cauchy–Binet, the principal Gram minor is a sum of squared minors of Y_i. For a row set E and order set T of equal size, such a squared minor has the form

$$e^{-\sum_E \lambda_{r,s}} \prod_{j\in T} \frac{e^{2\ell j}}{R^{2j}} \big(\det[(r)_{\underline{j}}]_{E,T}\big)^2 q^{\sum_E (r+s) - \sum_T j}.$$

If the determinant is nonzero, at least one term in its expansion is nonzero. That term matches each j to a row with $r \geq j$, so the exponent of q is nonnegative. All coefficients are nonnegative, proving the claim; the value at $q = 0$ follows by continuity. □

Thus a node can be moved outward to a radius bound before estimating its selected Gram minor. This assertion concerns principal minors, not monotonicity of individual eigenvalues. It is exactly the form needed to replace the ordered nodes by a finite set of radial bounds.

4.2. **Why negative-binomial weights help.** Fix parameters $a > 0$, $0 < v < 1$ and $\beta < 0$, and let $b = aR$. We use

$$e^{-\lambda_{r,s}} = \frac{(b)_r}{r!} v^r e^{-\beta s}, \tag{4.2}$$

where $(b)_r = b(b+1)\cdots(b+r-1)$. These are negative-binomial weights in the r variable. They are useful because their moments against falling factorials can be computed explicitly.

The triangle allows a better estimate than completing the rows to a rectangle. Write $x_+ = \max(0, x)$. At a node of squared radius $q > 0$, put

$$\delta_q = (\log q - \beta)_+, \qquad v_q = v e^{-\delta_q/K}, \qquad w = v_q q, \qquad \mathcal{B} = S e^{S\delta_q}.$$

For a fixed $r < R$, the permitted s satisfy $0 \leq s < S - r/K$. Hence

$$\sum_{0\leq s<S-r/K} (q e^{-\beta})^s \leq S e^{(S-r/K)\delta_q}. \tag{4.3}$$

Multiplication by $e^{-r\delta_q/K}$ replaces v by v_q. We may then extend the r sum to all nonnegative integers: the added row outer products are positive semidefinite. After removing the node angle by diagonal unitary conjugation, G_i is bounded above in the positive semidefinite order by the $J \times J$ matrix

$$\mathcal{G}_{jk} = \mathcal{B} \left(\frac{e^{\ell}}{R\sqrt{q}}\right)^{j+k} \sum_{r\geq 0} \frac{(b)_r}{r!} w^r (r)_{\underline{j}} (r)_{\underline{k}}, \qquad 0 \leq j, k < J. \tag{4.4}$$

Every principal minor of $\mathcal{G}$ majorizes the corresponding principal minor of G_i. The formulas at $q = 0$ are understood by continuity. In the finite radial partition used below all upper radii are positive.

Here is the moment calculation behind the choice of weights. Normalize $(b)_r w^r / r!$ to have total mass one. The resulting random integer r satisfies

$$\mathbb{E}[(1+x)^r (1+y)^r] = \left(\frac{1-w}{1-w(1+x)(1+y)}\right)^b. \tag{4.5}$$

Factoring the denominator as

$$(1-w)\big((1-\alpha x)(1-\alpha y) - \delta_0 xy\big), \qquad \alpha = \frac{w}{1-w}, \quad \delta_0 = \frac{w}{(1-w)^2},$$

and expanding in x and y gives a triangular orthogonal factorization. Let P_l be the monic polynomials orthogonal for this normalized negative-binomial measure. These are the Meixner

polynomials; their formulas are

$$\|P_l\|^2 = l!(b)_l \frac{w^l}{(1-w)^{2l}}, \tag{4.6}$$

$$(r)_{\underline{j}} = \sum_{l=0}^{j} \binom{j}{l} (b+l)_{j-l} \left(\frac{w}{1-w}\right)^{j-l} P_l(r). \tag{4.7}$$

These formulas are standard [3, Sections 18.19 and 18.23]; they can also be checked from (4.5).

Choose a column scale $\tau > 0$ and set

$$\nu_j = \log(j!) + \log((b)_j) + 2j(\ell - \log(R\tau)). \tag{4.8}$$

After dividing column j by $e^{\nu_j/2}$, the factorization (4.6)–(4.7) identifies the completed matrix factor with $\sqrt{\mathcal{B}(1-w)^{-b}}$ times the matrix of composition by

$$\phi(z) = c_0 + dz, \qquad c_0 = \frac{\tau v_q \sqrt{q}}{1-w}, \qquad d = \frac{\tau\sqrt{v_q}}{1-w}, \tag{4.9}$$

on polynomials of degree less than J. The Hilbert space has orthonormal monomials $\sqrt{(b)_j/j!}\, z^j$, or equivalently reproducing kernel $(1-z\bar{w})^{-b}$. For example, the coefficient in row l, column j of composition is

$$\binom{j}{l} \sqrt{\frac{(b)_j l!}{j!(b)_l}} c_0^{j-l} d^l,$$

which is precisely the normalized factor obtained above. This converts a large derivative Gram matrix into a problem about an affine map of the disk.

4.3. **Two bounds for singular products.** We state the two estimates used in the spectral part of the computation. We prove the first here and give the second proof in Appendix C.1.

Lemma 4.2 (Affine composition)**.** *Suppose $c_0 \geq 0$, $d > 0$ and $c_0 + d < 1$. Set*

$$U = c_0 d, \quad V = d^2, \quad Z = 1 - c_0^2, \qquad h = \frac{2U}{Z - V + \sqrt{(1-(c_0+d)^2)(1-(c_0-d)^2)}},$$

$$\Gamma = (Z - Uh)^{-1}, \qquad \xi^2 = \frac{V + Uh}{Z - Uh}.$$

The squared singular values of composition by $c_0 + dz$ on the space with kernel $(1-z\bar{w})^{-b}$ are $\Gamma^b \xi^{2l}$, $l \geq 0$.

Proof. One has $0 \leq h < 1$, $0 < \xi < 1$, and $Uh^2 - (Z - V)h + U = 0$. The kernel of the product of the composition operator with its adjoint is

$$[Z - U(z + \bar{w}) - Vz\bar{w}]^{-b}.$$

The disk automorphism $(z-h)/(1-hz)$, accompanied by the multiplier $(1-h^2)^{b/2}/(1-hz)^b$, induces a unitary operator, as is checked by substituting in the reproducing kernel. Conjugating the displayed kernel by this unitary and using the quadratic equation for h gives Γ^b times the kernel of the dilation operator $f(z) \mapsto f(\xi^2 z)$. Its eigenvalues on the monomials are ξ^{2l}. Taking square roots gives the singular values of the original operator. □

Compression to polynomials of degree less than J cannot increase singular values. If m orders are selected at a node, let $\nu_m^*(\tau)$ be the sum of the largest m numbers among $\nu_0, \ldots, \nu_{J-1}$. Restoring the column scales and the scalar mass gives the bound

$$F_q^{\mathrm{f}}(m) = \nu_m^*(\tau) + m \log \mathcal{B} + bm \log \frac{\Gamma}{1-w} + m(m-1) \log \xi \tag{4.10}$$

for its logarithmic Gram determinant. This applies to every selection of m orders. The term $m(m-1)\log\xi$ is negative and quadratic in m. It records the geometric decay of the singular values, a saving lost when all m singular values are bounded by the largest one. This explains why selecting many columns at a node can improve the estimate. Conveniently, one may choose $0<\xi<1$ first and then set

$$\Gamma=\frac{1+\xi^2+\sqrt{(1-\xi^2)^2+4w\xi^2}}{2},\qquad \tau=\frac{\xi(1-w)}{\Gamma\sqrt{v_q}}.$$

These choices automatically give a contracting map in (4.9).

Near $m=J$, it can be better to estimate the complementary singular product. This allows an expanding finite affine map as well.

Lemma 4.3 (Complementary singular product)**.** *Let T_J be composition by c_0+dz on polynomials of degree less than J, where $c_0\ge 0$ and $d>0$. Put $k=J-m$. For $0<r_0<1$, define*

$$G_0=\frac{1+r_0^2+\sqrt{(1-r_0^2)^2+4c_0^2r_0^2}}{2},\qquad s_0=\frac{dr_0}{G_0}.$$

The logarithm of the product of the largest m squared singular values of T_J is at most

$$J(J-1)\log d+kb\log G_0+k(k-1)\log r_0-N\log s_0, \tag{4.11}$$

where $N=k(2J-k-1)$ for $s_0\le 1$, and $N=k(k-1)$ for $s_0\ge 1$.

Adding $\nu_m^*(\tau)+m\log\mathcal{B}-bm\log(1-w)$ to (4.11) bounds the same node minor as (4.10). We can take the minimum over either estimate and over any finite collection of its parameters. Each choice gives an upper bound for the original selected minor, so minimizing does not presume that a single parameter is optimal at every node.

4.4. **A bound summed over all order sets.** The preceding estimates bound a selected Gram minor through singular values. We can also bound it by summing all principal minors of the same size: each term is nonnegative, so the sum bounds any one of them, regardless of which derivative orders occur in the nonsingular minor. The explicit moments above give the following estimate for this sum. Its Schur-function proof is given in Appendix C.2. Let

$$M_{jk}(b,w)=\sum_{r\ge 0}\frac{(b)_r}{r!}w^r(r)_{\underline{j}}(r)_{\underline{k}},\qquad G_{jk}=\mathcal{B}\sigma_0^{j+k}M_{jk}(b,w),$$

where $b,\mathcal{B},\sigma_0>0$ and $0<w<1$. Put $c_*=\sigma_0\sqrt{w}/(1-w)$ and $z_*=(bc_*)^{-2}$. If $bc_*>1$, then, for $0\le m\le J$, with $a_*=J-m$ and $s_*=ma_*+m(m-1)/2$,

$$\sum_{\substack{T\subseteq\{0,\dots,J-1\}\\ |T|=m}}\det G[T,T]\le\mathcal{B}^m(1-w)^{-bm}c_*^{2s_*}\prod_{i=0}^{m-1}i!(b)_{2a_*+i}\,(1-z_*)^{-m^2}. \tag{4.12}$$

The empty product is one. In our application $G=\mathcal{G}$ from (4.4), with $\sigma_0=e^{\ell}/(R\sqrt{q})$. Taking the logarithm of the right side of (4.12) therefore gives a third upper bound for every selected node minor.

The reason Schur functions appear can be described without using them in the rest of the proof. The Meixner factorization and Cauchy–Binet first bound a selected minor by a gamma moment determinant. If its order set is $j_0<\cdots<j_{m-1}$, its deviation from consecutive orders is a partition. The gamma determinant is bounded by the consecutive-order value times the square of the corresponding Schur dimension. Summing these dimensions over all partitions uses the Cauchy identity and produces $(1-z_*)^{-m^2}$. It is the sum over all order sets, rather than an assertion that

one consecutive set is largest, that makes (4.12) valid. The numerical estimate takes its minimum with the two singular-product bounds.

4.5. **Combining nodes with an occupancy constraint.** Let $F_q(m)$ be any common upper bound for the logarithmic determinant of an m-column minor at squared radius q, obtained as above. For a real parameter $a_0 > 0$, choose $\Phi(q) \geq 0$ such that

$$F_q(m) \leq a_0 R m + R^2 \Phi(q) \qquad (0 \leq m \leq J). \tag{4.13}$$

Any function Φ satisfying this inequality will suffice. Summing over nodes and using $\sum_i |C_i| = M$, (4.1) gives

$$\frac{1}{W} \log |\det B_C(\ell)|^2 \leq U_{\mathrm{row}} + \frac{3a_0(S+1)}{2S} + \frac{3}{S} \sum_{i=1}^{n} \Phi(q_i), \qquad U_{\mathrm{row}} = \frac{1}{W} \sum_{\Lambda} \lambda_{r,s}. \tag{4.14}$$

This elementary use of a common slope incorporates the total number of selected columns. Bounding each node independently at its worst occupancy would discard that information.

The disk estimate supplies

$$q_i \leq Q(i/n), \qquad Q(t) = \left(\frac{\sqrt{t} + \eta}{1 + \eta} \right)^2, \qquad \eta = \frac{\sqrt{D+4}/4}{\sqrt{n\sqrt{D}/(2\pi)}}.$$

Partition i/n into N_t intervals. For the kth interval, first move all its nodes outward to squared radius $Q(k/N_t)$ using Lemma 4.1, and then apply (4.13) at that radius. This needs no monotonicity assumption on the chosen function Φ. Each interval contains at most $n/N_t + 1$ nodes. Substituting $n = R(S+1)/(2\theta J)$ gives

$$U_{\mathrm{var}} = U_{\mathrm{row}} + \frac{3a_0(S+1)}{2S} + 3 \left(\frac{1 + 1/S}{2\theta (J/R) N_t} + \frac{1}{S} \right) \sum_{k=1}^{N_t} \Phi(Q(k/N_t)), \qquad \theta = \frac{M}{nJ}. \tag{4.15}$$

The nonnegativity of Φ justifies using upper bounds for the node counts. The signed row sum U_{row} remains outside that estimate.

For the finite comparisons below we take

$$(a, v, \beta, a_0) = \left(\frac{1}{5}, \frac{7}{10}, -\frac{4}{5}, \frac{3}{50} \right), \qquad (N_t, N_z, N_m) = (128, 256, 512). \tag{4.16}$$

The latter two mesh sizes are used for column-order capacity and occupancy, respectively. Whole-cell upper bounds are taken before choosing the minimum of the three estimates; endpoints alone would not bound an entire cell. Appendix C proves these finite bounds. Substitution into (4.15), with directed rounding, supplies U_{var} in the determinant comparison.

5. Nonvanishing with fewer nodes

The last algebraic input is full row rank of the matrix V. A dimension count requires $nJ \geq M$, but it does not prove that the rows are independent. The criterion below comes close to that count and is sufficient for all the finite comparisons between 499 and 875. The proof uses a divided-Frobenius basis to retain information lost by direct reduction modulo a prime.

Direct reduction modulo an inert prime loses information: the polynomial $z^p - w$ vanishes at every reduced node. The remedy is to divide out the powers of p whose presence we already know. After this division, a hypothetical row relation gives a polynomial in two variables with many prescribed local zeros. A differential operator converts those conditions into too many zeros of a nonzero polynomial in one variable. The two main points are to prove that this last polynomial is nonzero and to compare its degree with its orders of vanishing.

Theorem 5.1 (Sufficient rank criterion)**.** *Let p be an inert prime, and suppose that the nodes $z_1, \dots, z_n$ have distinct residues modulo $p\mathcal{O}_D$. Let $R = KS$ and*

$$K < p < R \le p^2, \qquad 1 \le J \le R.$$

If $n \ge 2S$ and

$$n\left(J - \frac{RJ}{p^2} - \frac{R}{p}\right) > \frac{(R+p)^2}{2K}, \tag{5.1}$$

then the matrix V in (2.2) has full row rank.

The leading terms of (5.1) are $nJ > R^2/(2K)$, close to the necessary inequality $nJ \ge M = R(S+1)/2$. The errors are small when p/R and R/p^2 are small. For our application the choice of p must also allow distinct node residues and guarantee that at least one candidate prime is inert. These requirements will be checked together in Section 6.

5.1. **Dividing out the Frobenius factors.** Put $g = z^p - w$ and $d = \lfloor (R-1)/p \rfloor < p$. The integral basis change (2.4) gives the rows

$$g^m z^a w^s, \qquad 0 \le m \le d, \quad 0 \le a < p, \quad a + Ks < R - mp. \tag{5.2}$$

At node i write

$$g(z_i + t, \bar{z}_i) = pU_i(t) + t^p, \quad U_i(t) = \frac{z_i^p - \bar{z}_i}{p} + \sum_{v=1}^{p-1} \frac{1}{p}\binom{p}{v} z_i^{p-v} t^v \in \mathcal{O}_D[t].$$

Direct reduction would replace $(pU_i(t) + t^p)^m$ by t^{mp}, discarding U_i and making layer m vanish in every column of order less than mp. Instead, divide each row in layer m by p^m and multiply each column of order $j = qp + v$, $0 \le v < p$, by p^q. Call the resulting matrix $\widetilde{V}$. Its entries are integral. Indeed, in the binomial expansion of $(pU_i + t^p)^m$, the term indexed by q' can contribute to order j only when $q' \le q$; after rescaling its coefficient contains $p^{q-q'}$. Modulo p, only the term $q' = q$ survives.

Write $k = \mathcal{O}_D/p\mathcal{O}_D$, $\xi_i = z_i \bmod p$, and $u_i(z) = (U_i \bmod p)(z - \xi_i)$. Thus, when $q \le m$, the reduced entry is

$$\binom{m}{q}[t^v](u_i(\xi_i + t)^{m-q}(\xi_i + t)^a \xi_i^{ps}). \tag{5.3}$$

When $q > m$, all terms are divisible by p, and the reduced entry is zero. The basis change and all scalings are invertible in characteristic zero. It therefore suffices to prove that $\widetilde{V} \bmod p$ has full row rank over k. These scalings are used only for nonvanishing; the determinant comparison continues to concern the original matrix V.

Suppose, to the contrary, that the reduced rows satisfy a nontrivial relation with coefficients $c_{m,a,s} \in k$. Package that relation as

$$F(z,u) = \sum_{m=0}^{d} H_m(z) u^m, \qquad H_m(z) = \sum_{a,s} c_{m,a,s} z^{a+ps}. \tag{5.4}$$

Within each layer the exponents $a + ps$ are distinct, since $0 \le a < p$. Consequently $F \ne 0$. Write $e = \deg_u F \le d < p$. The triangular support gives

$$\deg H_m < E_m := \frac{p(R - mp)}{K}, \tag{5.5}$$

because $a + ps = (p/K)(a + Ks) - (p/K - 1)a < p(R - mp)/K$. The importance of this bound is its constant drop: $E_m - E_{m+1} = p^2/K > p$.

To express the local conditions on F, put

$$L_l = \min(p, \max(0, J - lp)).$$

There are exactly L_l derivative orders in the block $j = lp + v$. In characteristic p, $(\xi_i + t)^{ps} \equiv \xi_i^{ps} \pmod{t^p}$. We may therefore replace ξ_i^{ps} by $(\xi_i + t)^{ps}$ when taking any of the coefficients $[t^v]$ with $v < L_l$. Summing (5.3) against the relation coefficients gives

$$\operatorname{ord}_{z=\xi_i}\big((\partial_u^{[l]}F)(z, u_i(z))\big) \geq L_l \qquad (i = 1, \dots, n;\ l \geq 0). \tag{5.6}$$

Here $\partial_u^{[l]}$ denotes the Hasse derivative in u. Thus a relation among the scaled rows has become a family of vanishing conditions along polynomial curves $u = u_i(z)$.

5.2. **One differential operator for all nodes.** Differentiating $pU_i(t) = (z_i + t)^p - \bar{z}_i - t^p$ before reducing modulo p gives

$$u_i'(z) = z^{p-1} - (z - \xi_i)^{p-1}. \tag{5.7}$$

The first term is independent of the node, and the second vanishes to order $p - 1$ there. This suggests using the common operator

$$\mathcal{D} = \partial_z + z^{p-1}\partial_u$$

in place of differentiation along each separate curve. Its key property is

$$\mathcal{D}^p = -\partial_u \qquad \text{on } k(z)[u]. \tag{5.8}$$

Indeed, the pth iterate of a derivation in characteristic p is a derivation. It sends z to zero and u to $(p-1)! = -1$, which determines it on $k(z)[u]$.

Lemma 5.2. *The polynomials $F, \mathcal{D}F, \dots, \mathcal{D}^e F$ are linearly independent over $k(z)$. In particular*

$$W_F(z) = \det\big([u^m]\mathcal{D}^j F\big)_{0 \leq j, m \leq e} \tag{5.9}$$

is nonzero, and

$$\deg W_F \leq \sum_{m=0}^{e} E_m - \frac{e(e+1)}{2}. \tag{5.10}$$

Proof. If these polynomials were dependent, let $r \leq e$ be the first index of a dependence. Their span up to index $r - 1$ would be stable under $\mathcal{D}$: applying $\mathcal{D}$ to a rational coefficient differentiates that coefficient, and the image of the last spanning vector is in the same span by assumption. By (5.8) this space is also stable under ∂_u. It would then contain $F, \partial_u F, \dots, \partial_u^e F$. Since $e < p$, these have distinct u-degrees $e, e-1, \dots, 0$ and nonzero leading coefficients. They are independent, contradicting the dimension $r \leq e$.

For the degree bound, induction using

$$[u^m]\mathcal{D}G = ([u^m]G)' + (m+1)z^{p-1}[u^{m+1}]G$$

shows that $\deg([u^m]\mathcal{D}^j F) \leq E_m - j$. The first term lowers degree by one; the second has degree at most $E_{m+1} - j + p - 1 < E_m - j - 1$. In every determinant term each column index m and row index j occurs once. Adding these degree bounds proves (5.10). □

The use of $e < p$ here is essential: it prevents the successive u-derivatives from vanishing prematurely in characteristic p. The identity $\mathcal{D}^p = -\partial_u$ is what connects this elementary fact to differentiation in z.

5.3. **Counting zeros of the coefficient determinant.** Fix a node i. Translation from the basis $1, u, \dots, u^e$ to $1, u-u_i(z), \dots, (u-u_i(z))^e$ is unitriangular. It leaves W_F unchanged and replaces its entries by

$$B^{(i)}_{j,l}(z) = (\partial_u^{[l]} \mathcal{D}^j F)(z, u_i(z)).$$

We claim that

$$\operatorname{ord}_{\xi_i} B^{(i)}_{j,l} \geq \max(0, L_l - j). \tag{5.11}$$

To see this, let $\mathcal{D}_i = \partial_z + u_i'(z)\partial_u$, which differentiates along the curve. Equation (5.7) gives $\mathcal{D} - \mathcal{D}_i = (z-\xi_i)^{p-1}\partial_u$. An induction, in which each differentiation lowers the order of the error by at most one, gives

$$\mathcal{D}^j G \equiv \mathcal{D}_i^j G \pmod{(z-\xi_i)^{p-j} k[z,u]} \qquad (0 \leq j < p).$$

The Hasse derivatives in u commute with both operators: their coefficients depend only on z, and ordinary and Hasse u-derivatives commute. After specializing $u = u_i(z)$, the corresponding entry is therefore congruent to the jth ordinary derivative of $(\partial_u^{[l]} F)(z, u_i(z))$, modulo $(z-\xi_i)^{p-j}$. Its order is at least $\max(0, L_l - j)$ by (5.6). Since $L_l \leq p$, the error has at least that order as well. This proves the claim.

We can now finish the proof by factoring powers of $z-\xi_i$ out of rows or columns. Put $Q = \lfloor J/p \rfloor$. If $e < Q$, then $L_l = p$ for all $0 \leq l \leq e$. Every entry in row j of $B^{(i)}$ has order at least $p-j$, so

$$\operatorname{ord}_{\xi_i} W_F \geq \sum_{j=0}^{e} (p-j) = (e+1)(p-e/2) > (e+1)p/2.$$

The residues are distinct and $n \geq 2S = 2R/K$. Summing this bound over the nodes therefore gives more than $(e+1)pR/K$ zeros counted with multiplicity, whereas (5.10) bounds the degree by at most that number. This contradicts $W_F \neq 0$.

If $e \geq Q$, every entry in column l has order at least $\max(0, L_l - e)$. Factoring column by column and using $\sum_{l=0}^{Q} L_l = J$ gives

$$\operatorname{ord}_{\xi_i} W_F \geq \sum_{l=0}^{Q} (L_l - e) = J - (Q+1)e \geq J - \frac{RJ}{p^2} - \frac{R}{p}. \tag{5.12}$$

For the degree, write $R = dp + s_0$, where $1 \leq s_0 \leq p$. Then

$$2p \sum_{m=0}^{d} (R - mp) = (R - s_0 + p)(R + s_0) = R^2 + pR + s_0(p - s_0) < (R+p)^2.$$

All E_m are positive. Hence

$$\deg W_F \leq \sum_{m=0}^{e} E_m \leq \sum_{m=0}^{d} E_m < \frac{(R+p)^2}{2K}.$$

Summing (5.12) over the nodes and applying (5.1) again makes the total order of vanishing exceed the degree. Both cases are impossible. The reduced matrix has full row rank, and so does V, proving Theorem 5.1.

6. The numerical comparisons and the bound 498

We now assemble the proof. There are two parts to the numerical argument. A uniform estimate, supplemented by adjacent interval comparisons, excludes every $D \geq 875$. Below 875 the elementary reduction leaves sixteen possible values, each of which is excluded by a separate determinant comparison. We describe the coverage first, then give the parameters and the decisive calculation at 499.

6.1. **The range $D \geq 875$.** For this range we use the resultant rank criterion of Proposition D.3 and the radial size estimate of Proposition E.2. Appendix F applies these estimates on an unbounded interval and then on finitely many adjoining intervals below it. Set $D_0 = 5\,760\,000$, $X = (D-3)/4$, and $X_0 = (D_0 - 3)/4$. On the unbounded interval, it gives

$$G_{\mathrm{tail}}(D) > 0.00031743 + 0.127829 \log(X/X_0) \qquad (D \geq D_0), \tag{6.1}$$

where G_{tail} is the normalized difference between the arithmetic lower bound and the radial upper bound. This difference is positive throughout $D \geq D_0$, contradicting the determinant inequality.

The interval between 875 and D_0 is covered as follows. The gap bounds in Table 1 are rounded downward from the smallest lower endpoint in each family. Each comparison uses a closed interval, and successive intervals share an endpoint.

Range	Number of intervals	Lower gap bound
$[875, 3\,968\,539]$	28	0.00397338
$[3\,968\,539, 4\,894\,699]$	37	0.00032629
$[4\,894\,699, 5\,760\,000]$	36	0.00040940

TABLE 1. Interval comparisons covering $[875, D_0]$. The first family uses norm-form estimates, the second exact finite prime sums, and the third explicit prime-sum bounds of Rosser and Schoenfeld.

Thus (6.1) and Table 1 exclude all $D \geq 875$ under the class-number-one hypothesis.

6.2. **Parameters and rank for the remaining comparisons.** By Lemma 2.4, the values still possible above 498 are 499 and the fifteen primes in Table 3. For these comparisons we use $K = 43$ and a prime cutoff P. The other parameters are

$$S = \left\lfloor \frac{6P}{5K} \right\rfloor, \qquad R = KS, \qquad J = \lfloor R/8 \rfloor, \qquad M = R(S+1)/2.$$

For $D = 499$ we take $P = 312\,000\,000$ and $n = 35\,394\,400$. Table 2 gives all four parameter choices.

P	S	R	J	n
36 000 000	1 004 651	43 199 993	5 399 999	4 822 400
169 000 000	4 716 279	202 799 997	25 349 999	22 638 400
312 000 000	8 706 976	374 399 968	46 799 996	41 793 600
312 000 000	8 706 976	374 399 968	46 799 996	35 394 400

TABLE 2. Parameters for all determinant comparisons below 875. The last line is used on $[499, 503]$; the other lines at the individual discriminants listed in Table 3.

We first prove nonvanishing. There are 814 primes in $(A, B] = (160\,000, 170\,000]$. Lemma B.5 bounds the number of non-inert primes by a sum of square roots. This sum is less than 765, both on $[499, 503]$ and at each value in Table 3. Hence at least one of the 814 primes in the interval is inert.

Lemma 2.5 gives $2\rho_n < 28\,928$ for every comparison in Table 2, and $2\rho_n < 22\,492$ at $D = 499$. Both are below $A = 160\,000$, so Corollary 2.6 gives distinct residues modulo every prime in $(A, B]$. Moreover, $K < A < B < R < A^2$ in every case. The remaining rank conditions are $J \leq R$, $n \geq 2S$, and

$$n\left(J - \frac{RJ}{A^2} - \frac{R}{A}\right) - \frac{(R+B)^2}{2K} > 0. \tag{6.2}$$

At the smaller node count the difference in (6.2) exceeds 7.22×10^{11}. The reason for narrowing the prime interval can now be seen quantitatively. Writing $\theta = M/(nJ)$, inequality (6.2) is equivalent to

$$\theta < \frac{(1 + 1/S)(1 - R/A^2 - R/(AJ))}{(1 + B/R)^2}.$$

At the parameters for 499, the right side is approximately 0.984431 when $A = 160\,000$, just above the actual occupancy 0.983995. With $A = 150\,000$ it is approximately 0.982414, which is too small. The narrower interval improves the sufficient rank estimate while still containing more primes than the norm form can account for as non-inert.

At the other three parameter choices the respective differences exceed 4.12×10^{12}, 9.02×10^{13} and 2.95×10^{14}. Theorem 5.1 therefore proves full row rank in every case.

6.3. **The fifteen comparisons above 499.** We now evaluate Proposition 2.7, with the local arithmetic estimates of Proposition B.3 and the spectral bound (4.15). The spectral parameters and mesh sizes are fixed as in (4.16). Table 3 gives the result; each displayed lower bound is rounded downward. The calculation uses outward rounding. Each entry is positive and contradicts

D	$P/10^6$	Lower gap bound	D	$P/10^6$	Lower gap bound
523	312	0.01913120	683	36	0.24689369
547	312	0.06770128	691	36	0.25866104
563	169	0.09816857	739	36	0.33084083
571	169	0.10589325	787	36	0.39774315
587	36	0.08275416	811	36	0.42960657
619	36	0.13882635	827	36	0.45047644
643	36	0.17968979	859	36	0.48970182
659	36	0.20911703			

TABLE 3. Positive lower bounds for $G = \mathcal{A} - \mathcal{U}$ at all fifteen primes $D \equiv 3 \pmod 8$ in $(499, 875)$.

(2.17). Thus every value in the table is excluded.

6.4. **The decisive comparison at 499.** At $D = 499$ we retain $P = 312\,000\,000$, $K = 43$ and the same R, S, J as in the preceding comparison, but reduce the node count to $n = 35\,394\,400$. The arithmetic and size calculations are performed on the whole interval $[499, 503]$, using the same parameters throughout. Only 499 needs to be excluded: $503 \equiv 7 \pmod 8$ is already impossible by Lemma 2.4.

Table 4 separates the three contributions to the comparison. The common term is

$$U_{\mathrm{common}} = \frac{W_1 + W_2}{W} \log \rho + \frac{Mc + 2M \log M}{W},$$

so that $G = \mathcal{A} - U_{\mathrm{common}} - U_{\mathrm{var}}$. The displayed arithmetic numbers are lower bounds and the size numbers are upper bounds, rounded in the safe direction. The first numerical column uses the larger node count, with all other parameters fixed.

The arithmetic decomposition (3.7) at the smaller node count is, approximately,

$$\mathcal{A} \approx 15.49500 - 6.91595 + 1.37599 + 0.03990 \approx 9.99494.$$

The last two contributions are the two- and three-factor corrections. Together they restore about 1.416 to the arithmetic lower bound, far more than the final margin; even the three-factor term, about 0.040, is comparable to that margin.

	Larger node count	Smaller node count
n	41 793 600	35 394 400
$\theta = M/(nJ)$, approximately	0.83333113	0.98399487
Arithmetic lower bound $\mathcal{A}$	9.99153787	9.99493844
Upper bound for U_{common}	9.71143220	9.62002864
Upper bound for U_{var}	0.40353803	0.32371122

TABLE 4. The comparison on $[499, 503]$ with $M = 1\,629\,945\,955\,088\,368$. Only the occupancy row uses approximate decimals; the other decimal entries are one-sided bounds.

The change in the gap is about 0.1746. Of this, 0.0914 comes from the common term, 0.0798 from the variable spectral term, and 0.0034 from the arithmetic lower bound. The common-term improvement is close to the radius estimate 0.085 in Section 2.4. The variable term benefits from the higher occupancy and the changed residual scale ℓ.

The complete directed calculation gives the following enclosure for the computed gap bound at the smaller node count, rounded outward here:

$$G \in [0.05119859,\ 0.05120060]. \tag{6.3}$$

In particular $G > 0.05119859$, contradicting the determinant comparison. For the larger node count the corresponding enclosure is

$$[-0.12343235,\ -0.12343034].$$

Proof of Theorem 1.1. Suppose that $h(-D) = 1$ and $D > 498$. Lemma 2.4 makes D a prime congruent to 3 modulo 8. Equation (6.1) and Table 1 exclude $D \geq 875$. Table 3 excludes all the remaining possibilities greater than 499, and (6.3) excludes 499 itself. These cases exhaust all $D > 498$. □

At 499, the occupancy is already about 0.984, so the number of nodes is only about 1.6% above the dimension-count minimum M/J. There is therefore little room to reduce the radius further by improving rank alone: in the radius approximation of Section 2.4, reaching the dimension count exactly would save only about 0.008 in the common term. Sharper arithmetic or spectral estimates are natural candidates for further progress.

The class number one hypothesis is used to turn non-inert prime ideals into representations by the norm form. A comparable estimate for general Landau–Siegel zeros would require a substitute for this arithmetic input.

APPENDIX A. LOCAL DIVISIBILITY AND FINITE ROW SUMS

The local estimates have two separate ingredients. A change of polynomial basis makes divisibility visible in each row. A counting argument then bounds how much of that divisibility can be lost to the derivative orders in a selected minor. We give both arguments here, including the finite corrections needed for the small primes.

Throughout this appendix, K, S, J, n are positive integers, $R = KS$, and

$$\Lambda = \{(r, s) \in \mathbb{Z}^2_{\geq 0} : r + Ks < R\}, \qquad M = \frac{R(S+1)}{2}, \qquad W = \frac{R^2 S}{3}, \qquad \theta = \frac{M}{nJ} \leq 1.$$

The nodes $z_1, \ldots, z_n$ may be any elements of $\mathcal{O}_D$. Columns are indexed by (i, j) with $1 \leq i \leq n$ and $0 \leq j < J$, and C denotes any selection of M columns. Thus each derivative order occurs in C at most n times. The determinant Δ_C is formed with the Hasse derivatives in (2.2); differentiation is in z, with w held fixed.

A.1. **Passing from entries to a determinant.** The following elementary observation allows different derivative estimates to be used on different sets of rows.

Lemma A.1. *Let $A = (a_{rc})$ be a square matrix over a discretely valued field, and let v be its valuation. Suppose that E is a set of rows and*

$$v(a_{rc}) \geq a_r - b_c - \mathbf{1}_{r\in E} d_c, \qquad d_c \geq 0.$$

Then

$$v(\det A) \geq \sum_r a_r - \sum_c b_c - \max_{\substack{I\subseteq C \\ |I|=|E|}} \sum_{c\in I} d_c.$$

An integral unitriangular change of rows does not change the determinant. If columns of order j have cost $d(j)$, each order is available at most n times, and $d(j)$ is nondecreasing, the maximum cost of m columns is bounded by filling the orders $J-1, J-2, \ldots$ in that order, with capacity n at each order. The minimum of a nondecreasing cost is bounded from below by filling orders $0, 1, \ldots$ in the same way.

Proof. Each term of the determinant uses every row and every column once. The columns paired with rows in E form a set of size $|E|$, so the displayed entry estimate gives the asserted lower bound for every determinant term. The valuation of a sum is at least the minimum valuation of its terms. A unitriangular row transformation has determinant one. Finally, if a selected column has smaller cost than an available column, exchanging them cannot decrease the total cost. Repeating this exchange proves the capacity assertions, and reversing the exchanges proves the assertion about minima. □

A.2. **Two finite row sums.** For integers $N \geq 0$ and $m \geq 1$, define

$$T(N,m) = \sum_{s=0}^{N-1} (N-s) \lfloor s/m \rfloor .$$

An empty sum is zero. Expressing $\lfloor s/m \rfloor$ as the number of positive multiples of m not exceeding s, and interchanging the two finite sums, gives

$$T(N,m) = \frac{1}{2} \sum_{\substack{a\geq 1 \\ am<N}} (N-am)(N-am+1). \tag{A.1}$$

There are $K(S-s)$ rows with a given second coordinate s. Consequently

$$\sum_{(r,s)\in\Lambda} \sum_{a\geq 1} \left\lfloor s/q^{2a} \right\rfloor = K \sum_{a\geq 1} T(S, q^{2a}).$$

In particular, this sum is finite even when written with no upper limit.

The corresponding sum in the first coordinate is conveniently bounded by

$$S_r(R,K,x) = \sum_{a=1}^{\lfloor R/x \rfloor} (R-ax)(R+K-ax), \qquad x > 0. \tag{A.2}$$

For an integer $q \geq 1$, put $Q = \lfloor R/q \rfloor$ and $c_a = aq \bmod K$. Then the exact identity is

$$2K \sum_{\Lambda} \lfloor r/q \rfloor = S_r(R,K,q) + \sum_{a=1}^{Q} c_a(K-c_a) \geq S_r(R,K,q). \tag{A.3}$$

To see this, count the rows with $r \geq aq$. Write $aq = Kb + c$, where $0 \leq c < K$, and put $u = S - b$. For each $s = 0, \ldots, u-1$ there are $K(u-s) - c$ such rows. Their number is $Ku(u+1)/2 - cu$. Multiplication by $2K$ gives $(R-aq)(R+K-aq) + c(K-c)$. Summing over a proves (A.3).

We will also need the shape of this weight as its argument varies. On an interval on which $Q = \lfloor R/x \rfloor$ is fixed, summing a and a^2 gives

$$S_r(R,K,x) = QR(R+K) - \frac{Q(Q+1)(2R+K)}{2}x + \frac{Q(Q+1)(2Q+1)}{6}x^2. \tag{A.4}$$

At an endpoint $x = R/Q$, the summand that disappears is zero, so S_r is continuous. If $u = R - Qx$, differentiation of this polynomial yields

$$-\frac{d}{dx}(xS_r(R,K,x)) = Q\left(\frac{Q-1}{2}x^2 + u(2x-u) + K(x-u)\right).$$

For $0 < x < R$ we have $Q \geq 1$ and $0 \leq u < x$, and the right side is positive. Since $\log x/x$ is positive and decreasing for $x \geq 3$, it follows that

$$x \longmapsto \frac{S_r(R,K,x)\log x}{KW} \quad \text{is decreasing for } 3 \leq x < R. \tag{A.5}$$

It is zero for $x \geq R$. This is the monotonicity used when replacing a possible prime norm by a lower endpoint for that norm.

For the unbounded range, an explicit lower bound is more useful than a piecewise polynomial. Set $\vartheta = \{R/q\}$ and substitute $Q = R/q - \vartheta$ into (A.4). Expanding gives

$$\begin{aligned} S_r(R,K,q) = {} & \frac{R^3}{3q} - \frac{R^2}{2} + \frac{KR^2}{2q} + \frac{qR}{6} - \frac{KR}{2} \\ & - \frac{q^2}{6}\vartheta(1-\vartheta)(1-2\vartheta) + \frac{Kq}{2}\vartheta(1-\vartheta). \end{aligned}$$

The last term is nonnegative, and $|\vartheta(1-\vartheta)(1-2\vartheta)| \leq 1/(6\sqrt{3})$. Using $KW = R^3/3$, we obtain

$$\frac{S_r(R,K,q)}{KW} \geq \frac{1}{q} - \frac{3}{2R} + \frac{3K}{2Rq} + \frac{q}{2R^2} - \frac{3K}{2R^2} - \frac{q^2}{12\sqrt{3}R^3}. \tag{A.6}$$

The cubic expression in ϑ is the usual third Bernoulli-polynomial remainder; the displayed calculation fixes both its sign and its constant.

A.3. **The basis for a small inert prime.** Fix an inert prime $q < K$, and write

$$h = h_q(r,s) = \min\left(\lfloor r/q \rfloor, \lfloor (R-1-r-Ks)/(K-q) \rfloor\right), \qquad \tau_q(s) = \sum_{a \geq 1} \left\lfloor s/q^{2a} \right\rfloor.$$

The first bound on h permits h copies of z^q to be replaced by w. The second ensures that the resulting monomials stay inside Λ. For fixed s put $t = S - s$. The inequalities $h \geq a$ say exactly $qa \leq r < Kt - (K-q)a$, so there are $K(t-a)$ such rows when $a < t$, and none otherwise. Summing these superlevel counts gives

$$2\sum_{\Lambda} h_q(r,s) = \frac{R(S^2-1)}{3} = W_2. \tag{A.7}$$

We first describe the rows on which the two derivative estimates will be used. In the part $0 \leq r < q(S-s)$, one has $h = \lfloor r/q \rfloor$, so $0 \leq r - qh < q$. In the remaining part, write

$$R - 1 - r - Ks = (K-q)h + e, \qquad 0 \leq e < K - q.$$

Here h is the second member of the minimum. If $d = S - s - 1 - h$ and $c = K - 1 - e$, then

$$r - qh = Kd + c, \qquad 0 \leq d < S, \quad q \leq c < K. \tag{A.8}$$

Conversely, for any such d, c, the values

$$0 \leq s < S - d, \qquad h = S - d - 1 - s, \qquad r = qh + Kd + c$$

give exactly these rows. Thus for each pair (d,c) there is a strip of length $t = S - d$, and there are $K - q$ strips of every length $1, \dots, S$. The number

$$m = \left\lfloor (r - qh)/q^2 \right\rfloor = \left\lfloor (Kd + c)/q^2 \right\rfloor$$

is constant on a strip. This constancy is what makes the finite estimate below possible.

Choose an integer L with $0 \le L \le S$, and let E_L consist of the strips of length at most L. In particular,

$$m_L := |E_L| = \frac{(K-q)L(L+1)}{2}, \qquad \varpi_L := \frac{m_L}{M}. \tag{A.9}$$

On these rows we will use all available factors $w^q - z$. On the other rows we keep the tower in w, which has no derivative cost.

Let T_a be the inert tower from (3.3). Writing $s = \sum_{a \ge 0} d_a q^{2a}$, put

$$B_s(w) = \prod_{a \ge 1} T_a(w)^{d_a}, \qquad u(s) = \lfloor s/q \rfloor \bmod q, \qquad r_0 = r - qh - q^2 m.$$

Thus B_s is monic of degree $q^2 \lfloor s/q^2 \rfloor$ and $v_q(B_s(\bar{z}_i)) \ge \tau_q(s)$. Define

$$P_{r,s}(z,w) = z^{r_0}(z^{q^2} - z)^m (z^q - w)^h w^{s \bmod q} \begin{cases} (w^q - z)^{\lfloor s/q \rfloor}, & (r,s) \in E_L, \\ (w^q - z)^{u(s)} B_s(w), & (r,s) \notin E_L. \end{cases} \tag{A.10}$$

In both cases the leading choice from the factors is $z^r w^s$, with coefficient one.

Here are the two checks needed to use these polynomials as row operations. First, replacing z^q by w increases the weight $r + Ks$ by $K - q$, but there are at most h such replacements. Their total increase is at most $R - 1 - r - Ks$. Every other replacement in (A.10) decreases the weight. All monomials therefore remain in Λ. Second, order monomials lexicographically by $(qr + s, s)$. Replacing z^{q^2} by z, or z^q by w, strictly decreases the first coordinate. Replacing w^q by z preserves the first coordinate and decreases the second. A lower term of B_s decreases the first coordinate. Hence the change from $z^r w^s$ to $P_{r,s}$ is integral and unitriangular. It preserves every selected minor.

A.4. **Valuations and derivative costs.** At a node, Frobenius and conjugation agree modulo an inert prime. Taylor expansion in z consequently has the form

$$(z_i + t)^q - \bar{z}_i = qU(t) + t^q, \qquad \bar{z}_i^q - (z_i + t) = qV - t,$$
$$(z_i + t)^{q^2} - (z_i + t) = qH(t) - t + t^{q^2},$$

with $U, H \in \mathcal{O}_D[t]$ and $V \in \mathcal{O}_D$. Let a denote the combined number of factors of the latter two types: $a = m + \lfloor s/q \rfloor$ on E_L, and $a = m + u(s)$ off E_L.

To lose a factor of q from one of the h factors of the first type, a Taylor term must spend at least q degrees in t. To lose one from any of the other a factors, it must spend at least one degree. If a_1 and a_2 are the respective numbers of such choices in a term of degree j, then $qa_1 + a_2 \le j$ and $a_2 \le a$. Its valuation is at least $h + a - a_1 - a_2$. These two inequalities imply, respectively,

$$h + a - a_1 - a_2 \ge h + a - j, \qquad h + a - a_1 - a_2 \ge h - \lfloor (j-a)/q \rfloor .$$

All factors not yet counted have integral coefficients. Thus the entry valuation, before adding the tower contribution, is at least

$$\max\left(0, h + a - j, h - \lfloor (j-a)/q \rfloor\right). \tag{A.11}$$

Set $\delta_q(j) = j - \lfloor j/q \rfloor$. For a row outside E_L, the last member of (A.11), together with the tower, gives

$$v_q(\partial_z^{[j]} P_{r,s}(z_i, \bar{z}_i)) \ge h + \tau_q(s) + \lfloor (m + u(s))/q \rfloor - \lfloor j/q \rfloor . \tag{A.12}$$

For a row in E_L, the middle member gives instead

$$v_q(\partial_z^{[j]} P_{r,s}(z_i, \bar{z}_i)) \geq h + \tau_q(s) + \big(m + \lfloor s/q \rfloor - \tau_q(s)\big) - \lfloor j/q \rfloor - \delta_q(j). \tag{A.13}$$

Both estimates have the same baseline row contribution $h + \tau_q(s)$ and the same column cost $\lfloor j/q \rfloor$. The extra row contribution is nonnegative in both cases, since $\tau_q(s) \leq \lfloor s/q \rfloor$. Only rows in E_L incur the additional cost $\delta_q(j)$.

A.5. **Summing the extra row contribution.** For the rows outside the cut, the following finite form of the floor-sum identity controls an incomplete strip.

Lemma A.2. *For integers $m, t \geq 0$ and $q \geq 2$,*

$$\sum_{s=0}^{t-1} \lfloor (m + (\lfloor s/q \rfloor \bmod q))/q \rfloor \geq \frac{tm}{q} - \frac{q^2}{4}.$$

Proof. Write $m = aq + b$, with $0 \leq b < q$. On each complete block of q^2 consecutive values of s, the summand is a for $q(q-b)$ values and $a+1$ for qb values. Its average is exactly m/q. It remains to examine a final block of length $kq + v$, with $0 \leq k, v < q$. If $k < q - b$, its deficit from this average is $b(k + v/q) \leq b(q - b)$. If $k \geq q - b$, the deficit is $(q - b)(q - k - v/q) \leq b(q - b)$. Both bounds are at most $q^2/4$. The case $b = 0$ has zero deficit. □

We now sum over the strips in (A.8). Put

$$c_{K,q} = \frac{K + q - 1}{2} - (q^2 - 1).$$

For a fixed length t, summing $\lfloor (K(S-t) + c)/q^2 \rfloor \geq (K(S-t) + c - (q^2-1))/q^2$ over $c = q, \ldots, K-1$ gives

$$\sum_{c=q}^{K-1} \left\lfloor (K(S-t) + c)/q^2 \right\rfloor \geq \frac{K-q}{q^2} (K(S-t) + c_{K,q}). \tag{A.14}$$

The right side is allowed to be negative; the exact extra contribution is still nonnegative.

Define

$$A_q(L) = \frac{2(K-q)}{q^2} \sum_{t=1}^{L} t(K(S-t) + c_{K,q}),$$

$$B_q(L) = \frac{2(K-q)}{q^3} \sum_{t=L+1}^{S} t(K(S-t) + c_{K,q}),$$

$$E_q(L) = A_q(L) + B_q(L) - \frac{q^2}{2}(S-L)(K-q) + 2(K-q)\left(T(L,q) - \sum_{a \geq 1} T(L, q^{2a})\right).$$

Twice the sum of the extra row contributions in (A.12)–(A.13) is at least $E_q(L)$. Indeed, on each strip of length $t \leq L$, the contribution of m is tm; (A.14) sums these contributions to at least $A_q(L)/2$. The remaining contribution on those strips is

$$(K-q) \sum_{t=1}^{L} \sum_{s=0}^{t-1} (\lfloor s/q \rfloor - \tau_q(s)) = (K-q)\left(T(L,q) - \sum_{a \geq 1} T(L, q^{2a})\right).$$

Here a given s occurs $L - s$ times, explaining the function T.

For a strip of length $t > L$, Lemma A.2 bounds the extra contribution from below by $tm/q - q^2/4$. Summing the first term with (A.14) gives $B_q(L)/2$; there are $(S-L)(K-q)$ such strips. The rows in the remaining left part of the triangle have nonnegative extra contribution and may be omitted. These observations account for every term of $E_q(L)$. Since the exact total extra contribution is itself nonnegative, we have the stronger lower bound

$$2\sum_{\Lambda}(\text{extra row contribution}) \geq (E_q(L))_+ . \tag{A.15}$$

The positive part is taken before subtracting the extra derivative cost.

A.6. **The extra column cost and the retained weight.** The cost $\delta_q(j)$ is nondecreasing. Its prefix sum has a simple exact form:

$$\Sigma_q(B) := \sum_{j=0}^{B-1} \delta_q(j) = \frac{q-1}{2q}B^2 - \frac{r_B(q-r_B)}{2q}, \qquad r_B = B \bmod q. \tag{A.16}$$

This follows by writing $B = aq + r_B$ and summing the complete blocks of the floor function. It also holds for $B = 0$.

Write $m_L = nt + r$, where $0 \leq r < n$, and put $A = J - t$. By Lemma A.1, the extra derivative cost is at most

$$n(\Sigma_q(J) - \Sigma_q(A)) + r\delta_q(A-1).$$

The last term is omitted when $r = 0$. In particular, $A - 1$ is only used when $A \geq 1$. To estimate the remainder in this formula, rewrite it as

$$n\Sigma_q(J) - (n-r)\Sigma_q(A) - r\Sigma_q(A-1).$$

For $r = 0$ the term with $A - 1$ is again omitted. The remainder in $\Sigma_q(J)$ is nonpositive, whereas each of the other two remainders is bounded above in absolute value by $\lfloor q^2/4 \rfloor /(2q)$. The quadratic parts give

$$\frac{q-1}{q}\left(m_L J - \frac{m_L^2}{2n}\right) - \frac{q-1}{2q}r\left(1 - \frac{r}{n}\right).$$

Dropping the final nonpositive term proves the upper bound

$$\frac{q-1}{q}\left(m_L J - \frac{m_L^2}{2n}\right) + \mathbf{1}_{m_L>0}\frac{n\lfloor q^2/4 \rfloor}{2q}. \tag{A.17}$$

If $m_L = 0$, the exact cost is zero, so no remainder is charged.

Normalize twice this cost by W, and recall $m_L = \varpi_L M$ and $M/(nJ) = \theta$. Since $2M/W = 3(S+1)/(RS)$, the resulting expression is

$$C_q(L) = 3\left(1 + \frac{1}{S}\right)\varpi_L\left(1 - \frac{\varpi_L\theta}{2}\right)\frac{q-1}{q}\frac{J}{R} + \mathbf{1}_{L>0}\frac{n\lfloor q^2/4 \rfloor}{qW}.$$

We have now obtained every term of the small-prime formula.

Proposition A.3. *For an inert prime $q < K$, define*

$$w_{\text{keep}}(q) = \frac{W_2 + 2K\sum_{a\geq 1} T(S, q^{2a})}{W} + \max_{0\leq L\leq S}\left(\frac{(E_q(L))_+}{W} - C_q(L)\right). \tag{A.18}$$

Then every selected minor satisfies

$$v_q(\Delta_C) \geq \frac{W}{2}w_{\text{keep}}(q) - \sum_{(i,j)\in C}\lfloor j/q \rfloor .$$

The maximum is over integers L. Using any one admissible value of L in place of the maximum also gives a valid lower bound.

Proof. For the basis associated with L, the sum of the baseline row contributions is, by (A.7) and (A.1),

$$\sum_{\Lambda}(h+\tau_q(s)) = \frac{W_2}{2} + K\sum_{a\ge 1} T(S,q^{2a}).$$

Equation (A.15) bounds the extra row contribution from below, and (A.17) bounds its extra column cost from above. Apply Lemma A.1 to (A.12) and (A.13). Every choice of L gives a bound for the same determinant, because all the bases are unitriangular. Taking the best of these bounds proves the proposition. □

In particular, the argument includes $q = 2$ when 2 is inert in the maximal order. The factors $1 + 1/S$ and the last remainder in $C_q(L)$ are exact finite-size corrections.

A.7. **Split and ramified primes.** For completeness we give the local contributions at the other primes. They are needed when primes beyond the guaranteed inert range are retained. At a split prime, a tower in w gives a contribution with no derivative cost. At a ramified prime, the inert estimates remain valid in the prime ideal valuation, but this has half the norm contribution of the inert valuation.

Suppose first that q splits. Then $\mathcal{O}_D/q\mathcal{O}_D \simeq \mathbb{F}_q \times \mathbb{F}_q$, so $b^q - b$ is divisible by q for every $b \in \mathcal{O}_D$. Define monic polynomials

$$G_1(w) = w^q - w, \qquad G_{a+1}(w) = G_a(w)^q - q^{q^a-1}G_a(w), \qquad \mu_a = \frac{q^a-1}{q-1}.$$

They have degrees q^a, and $G_a(\bar z_i) \in q^{\mu_a}\mathcal{O}_D$. Indeed, if $G_a(\bar z_i) = q^{\mu_a}b$, the next value is $q^{q\mu_a}(b^q - b)$, divisible by $q^{q\mu_a+1} = q^{\mu_{a+1}}$. For the base-$q$ expansion $s = \sum_{a\ge 0} d_a q^a$, the polynomial

$$\widetilde{A}_s(w) = w^{d_0}\prod_{a\ge 1} G_a(w)^{d_a}$$

is monic of degree s and has values divisible by $q^{\sum_{a\ge 1}\lfloor s/q^a\rfloor}$. The basis $z^r\widetilde{A}_s(w)$ is unitriangular on Λ, and differentiating in z does not alter this divisibility. Hence

$$\Delta_C \in q^{K\sum_{a\ge 1} T(S,q^a)}\mathcal{O}_D. \tag{A.19}$$

Its contribution to $\log|\Delta_C|^2/W$ is at least

$$(e(q)+s(q))\log q, \qquad e(q) = \frac{2K}{W}\sum_{a\ge 1} T(S,q^{2a}), \quad s(q) = \frac{2K}{W}\sum_{a\ge 1} T(S,q^{2a-1}).$$

There is no first-power derivative debit in this estimate. When the arithmetic lower bound is organized with a common debit $\sum_C \lfloor j/q\rfloor$ for all primes, a split prime therefore permits that debit to be refunded. Its minimum over C is given by the capacity principle in Lemma A.1.

Now suppose that $q \ge K$ and $(q) = \mathfrak{q}^2$ is ramified. In our application the ramified prime is $q = D > K$. Normalize $v_\mathfrak{q}$ by $v_\mathfrak{q}(\mathfrak{q}) = 1$. Conjugation acts trivially on $\mathcal{O}_D/\mathfrak{q} \simeq \mathbb{F}_q$, so $z_i^q - \bar z_i$ and $\bar z_i^q - \bar z_i$ belong to $\mathfrak{q}$. A substitute for the inert tower is

$$\widehat{B}_s(w) = (w^q - w)^{q\lfloor s/q^2\rfloor}.$$

It is monic of degree $q^2\lfloor s/q^2\rfloor$, and

$$v_\mathfrak{q}(\widehat{B}_s(\bar z_i)) \ge q\left\lfloor s/q^2\right\rfloor \ge \tau_q(s).$$

For the last inequality, put $a = \lfloor s/q^2\rfloor$; then $\tau_q(s) = a + \lfloor a/q^2\rfloor + \lfloor a/q^4\rfloor + \cdots \le aq^2/(q^2-1) \le aq$.

Use the unitriangular basis

$$z^{r \bmod q}(z^q - w)^{\lfloor r/q\rfloor}w^{s \bmod q^2}\widehat{B}_s(w).$$

As in (3.4), its entries have valuation at least $\lfloor r/q \rfloor + \tau_q(s) - \lfloor j/q \rfloor$. Therefore

$$v_{\mathfrak{q}}(\Delta_C) \geq \frac{S_r(R,K,q)}{2K} + K \sum_{a \geq 1} T(S, q^{2a}) - \sum_{(i,j) \in C} \lfloor j/q \rfloor \,. \tag{A.20}$$

One may replace a negative right side by zero, since the entries are integral. Finally, an inert prime ideal has norm q^2, whereas a ramified prime ideal has norm q. The contribution of $v_{\mathfrak{q}}(\Delta_C)$ to $\log |\Delta_C|^2$ is consequently $v_{\mathfrak{q}}(\Delta_C) \log q$. This explains the factor of one half in the normalized ramified contribution used in the arithmetic comparison.

Appendix B. Norm representations and the arithmetic estimate

The local estimates of Appendix A treat each prime separately. To sum them, we begin by assigning the inert-prime contribution to every prime below a cutoff. We then subtract the excess assigned to split and ramified primes. Under class number one, these primes are represented by the norm form. Counting norm values gives an upper bound for the loss; removing composite values improves it. This appendix makes that argument precise and accounts for the derivative orders in the selected minor.

Throughout, $D > 163$ is prime, $D \equiv 3 \pmod 8$, and $h(-D) = 1$. Put $m_D = (D+1)/4$ and

$$N_D(x,y) = x^2 + xy + m_D y^2 = (x + y/2)^2 + Dy^2/4.$$

An element $x + y\omega$ is primitive if $\gcd(x,y) = 1$. This is equivalent to not being divisible in $\mathcal{O}_D$ by a rational integer greater than one. Since the units are $\{\pm 1\}$, every nonreal element has a representative modulo sign and conjugation with $y > 0$ and $2x + y \geq 0$.

B.1. **Prime and composite norm values.**

Lemma B.1. *Every split prime has exactly one primitive representative with $y > 0$ and $2x + y > 0$. The only primitive representative on the imaginary axis is $(-1, 2)$, of norm D. Every prime factor of a primitive norm is at least m_D.*

If a primitive norm has $r \geq 1$ distinct split prime factors, then it has 2^{r-1} representatives modulo sign and conjugation, whether or not the ramified prime also divides it. The ramified prime can occur only to the first power.

Proof. A prime ideal above a split prime q is principal. A generator has norm q, and its conjugate generates the other prime ideal. The four generators are related by sign and conjugation, giving the stated representative. They cannot lie on either axis. On the imaginary axis $2x + y = 0$; primitivity then gives $(x, y) = (-1, 2)$.

An inert prime dividing a norm divides the element itself in $\mathcal{O}_D$, so cannot divide a primitive norm. Every remaining prime is the norm of a nonreal element, and hence is at least m_D by Lemma 2.3. If $q = \pi\bar{\pi}$ is split, a primitive element cannot contain both π and $\bar{\pi}$. For each distinct split prime factor there is therefore a choice between its two conjugate prime factors, with the exponent already fixed by the norm. Conjugation pairs these 2^r choices. The ramified factor is associated to its conjugate, and its square is a rational integer times a unit. It supplies no further choice and cannot occur twice in a primitive element. □

The restriction on the prime factors gives a short way to make the necessary finite lists. The bound below concerns the cutoff for retained split-prime contributions; it need not hold at the larger cutoff used for the row estimate.

Lemma B.2. *Let $Q < m_D^4$. Every composite primitive norm at most Q is a product of two or three prime factors, counted with multiplicity. If $F \geq Q/m_D$ and $F < m_D^4$, all of its prime factors occur among the primitive elements of norm at most F after the reducible elements have been removed.*

To perform that removal, it suffices to form products of a primitive element of norm at most $\sqrt{F} < m_D^2$ and a primitive element of norm at most F/m_D.

Proof. Each prime factor is at least m_D, by Lemma B.1. Thus four factors would have product at least m_D^4, and any factor of a composite norm at most Q is at most Q/m_D.

If a primitive element of norm at most F is reducible, choose an irreducible factor of smallest norm. Its norm is at most $\sqrt{F}$, and the complementary factor has norm at most F/m_D. Both factors are primitive: any rational divisor of either would divide the original element. Conversely, a product of two nonunits is reducible. The indicated products therefore remove exactly the reducible elements. Under class number one the remaining nonunit elements have prime norm. □

Multiplication in these lists is ordinary multiplication in the order:

$$(x + y\omega)(x' + y'\omega) = (xx' - m_D yy') + (xy' + x'y + yy')\omega. \tag{B.1}$$

After multiplication one keeps the primitive products and identifies representatives related by sign or conjugation. Lemma B.1 gives their multiplicities. This provides both a complete list of composite values up to Q and, by subtraction from all primitive norm values, an exact list of the split primes up to Q.

B.2. **The loss at a non-inert prime.** Assume $2 \leq K < D/4$ and choose $P \geq \max(S, J-1)$. All primes below K are inert. For $x \geq K$, define

$$\begin{aligned} w(x) &= \frac{S_r(R, K, x)\log x}{KW}, \\ e(x) &= \frac{2K}{W}\sum_{a\geq 1} T(S, x^{2a}), \qquad s(x) = \frac{2K}{W}\sum_{a\geq 1} T(S, x^{2a-1}). \end{aligned} \tag{B.2}$$

The sums T and S_r are defined in Appendix A.2. The provisional row contribution is

$$\mathcal{A}_0 = \sum_{q<K} w_{\mathrm{keep}}(q)\log q + \sum_{K\leq q\leq P}(w(q) + e(q)\log q), \tag{B.3}$$

where all sums indexed by q are over primes. Its common derivative loss is $\sum_{q\leq j}\lfloor j/q\rfloor \log q$. The cutoff includes every prime appearing in this loss and every nonzero conjugate-variable term.

Choose a finite set $\mathcal{S}$ of normalized primitive representatives other than $(-1, 2)$, all of norm at most $J-1$, and let $\mathcal{Q}$ be the set of prime norms represented in $\mathcal{S}$. By Lemma B.1, these primes are split and each is represented once. When D varies over an interval, we keep the integer coordinate pairs in $\mathcal{S}$ fixed and require the norm bound throughout the interval; $\mathcal{Q}$ may then depend on D. We retain the first-power derivative contributions of $\mathcal{Q}$ individually. For other split primes, a bound independent of the selected columns is more convenient. Write $M = nt + r$, with $0 \leq r < n$, and put

$$C(x) = n\sum_{j=0}^{t-1}\lfloor j/x\rfloor + r\lfloor t/x\rfloor. \tag{B.4}$$

There are n columns of each order. Since $j \mapsto \lfloor j/x\rfloor$ is nondecreasing, every selection of M columns satisfies $\sum_C \lfloor j/x\rfloor \geq C(x)$.

At a split prime, the conjugate-variable estimate retains the even-power contribution $e(q)\log q$ and adds $s(q)\log q$. There is also no first-power derivative loss. Thus, with $\iota = 1$ when $q \in \mathcal{Q}$ and

$\iota = 0$ otherwise, an upper bound for the loss from the provisional row contribution is

$$L(q, \iota) = \left[w(q) - s(q) \log q - (1 - \iota) \frac{2C(q) \log q}{W} \right]_+ . \tag{B.5}$$

The positive part permits us to discard any improvement beyond the provisional contribution. When $\iota = 1$, the derivative contribution is instead removed from the common loss. The same contribution is therefore never used twice.

If $D \le P$, the ramified-prime estimate in Appendix A gives the loss bound

$$L_{\mathrm{ram}}(D) = \frac{1}{2} \left[w(D) + e(D) \log D - \frac{2C(D) \log D}{W} \right]_+ . \tag{B.6}$$

If $D > P$, this term is absent. In particular, a cutoff below $D/4$ requires no non-inert-prime correction at all.

B.3. **Summing the losses by the norm form.** Let $\mathcal{N}$ contain the normalized primitive representatives of norm in $[m_D, P]$, excluding $(-1, 2)$. For each representative α, put $\iota_\alpha = 1$ if $\alpha \in \mathcal{S}$ and $\iota_\alpha = 0$ otherwise. Define

$$L_{\mathrm{norm}} = \sum_{\alpha \in \mathcal{N}} L(N(\alpha), \iota_\alpha) + \mathbf{1}_{D \le P} L_{\mathrm{ram}}(D).$$

Every split prime at most P occurs once in this sum. Composite values add nonnegative terms, so L_{norm} is an upper bound for the actual loss. We may subtract any lower bound for these extra terms. Let C_2 and C_3 be such lower bounds, obtained from disjoint primitive representatives whose norms have respectively two and three prime factors. Their multiplicities are as in Lemma B.1.

Proposition B.3 (Arithmetic estimate)**.** *Suppose $R = KS$, $1 \le J \le R$, $M \le nJ$, $2 \le K < D/4$, and $P \ge \max(S, J - 1)$. Set*

$$\mathcal{A} = \mathcal{A}_0 - L_{\mathrm{norm}} + C_2 + C_3, \qquad b(j) = \sum_{q \le j} \lfloor j/q \rfloor \log q - \sum_{q \in \mathcal{Q}} \lfloor j/q \rfloor \log q. \tag{B.7}$$

Then every nonzero maximal minor of the integral interpolation matrix satisfies

$$W\mathcal{A} - 2 \sum_C b(j) \le \log |\Delta_C|^2.$$

The conclusion also holds if L_{norm} is replaced by an upper bound and C_2, C_3 by lower bounds. Additional composite corrections may be omitted.

Proof. Apply the inert-prime estimates of Appendix A provisionally to all primes up to P. For a split prime, replace its row contribution and derivative loss by the actual split-prime estimate. Equations (B.4) and (B.5) bound the cost of that replacement uniformly in C. Equation (B.6) does the same for the ramified prime. The product formula for $|\Delta_C|^2$ then gives the claimed inequality with the true prime losses. The norm sum bounds those losses from above, and $C_2 + C_3$ subtracts only terms contributed by composite representatives. This proves the result. □

This proposition applies both to cutoffs below the guaranteed inert range and to cutoffs beyond it. For the latter, the row sum may include norms greater than m_D^4. Only the complete removal of composite values in the set selecting $\mathcal{Q}$ requires the smaller cutoff in Lemma B.2.

B.4. **Bounds over an interval of discriminants.** The comparison on $[499, 503]$ and the first family of intervals above 875 use one estimate for several values of D. The norm form makes this possible because, for a fixed pair (x, y),

$$N_D(x, y) = N_{D_-}(x, y) + \frac{D - D_-}{4} y^2. \tag{B.8}$$

We may therefore enclose each norm by its endpoint values. The following observation identifies many factors without a separate test at every intermediate discriminant.

Lemma B.4. *Put $m_- = (D_- + 1)/4$. If a primitive pair with $y > 0$ satisfies $N_{D_-}(x, y) < m_-^2$, then $N_D(x, y) < m_D^2$ for every $D \geq D_-$. Consequently it is irreducible whenever $\mathcal{O}_D$ has class number one.*

Proof. The norm bound gives $y^2 < 4m_-^2/(4m_- - 1) < m_- + 1$, so $y^2 \leq m_-$. Writing $m_D = m_- + u$, equation (B.8) yields $N_D < m_-^2 + um_- \leq (m_- + u)^2$. A reducible primitive element has at least two prime factors, each of norm at least m_D. □

All possible primitive norm representatives are found by the finite conditions $N_{D_-}(x, y) \leq P$, $y > 0$ and $2x + y \geq 0$. Let $[a, b]$ enclose one such norm, restricting it to $[m_-, P]$ when bounding a possible prime. By (A.5), w is decreasing on $[3, R)$ and zero beyond; s and C are nonincreasing by their floor-sum definitions. These functions are nonnegative, so valid endpoint bounds are

$$L^+(a, b, \iota) = \left[w(a) - s(b) \log a - (1 - \iota) \frac{2C(b) \log a}{W} \right]_+, \tag{B.9}$$

$$L^-(a, b, \iota) = \left[w(b) - s(a) \log b - (1 - \iota) \frac{2C(a) \log b}{W} \right]_+. \tag{B.10}$$

At the ramified representative the analogous upper bound is

$$\frac{1}{2} \left[w(a) + e(a) \log b - \frac{2C(b) \log a}{W} \right]_+.$$

Summing (B.9) over all possible representatives bounds the loss from above. Products known to occur throughout the interval, with their multiplicities, can be subtracted using (B.10).

Products of norm intervals are handled as follows. Suppose the factors have norm intervals $[a_i, b_i]$, and put $A = \prod_i a_i$, $B = \prod_i b_i$. At each discriminant their product has norm in $[A, B]$, even if the order of the candidate norm values changes as D varies. If (X, Y) is its normalized representative at that discriminant, then $Y^2 \leq \lfloor 4B/D_- \rfloor$. Writing $\delta m = (D_+ - D_-)/4$, we obtain

$$N_{D_+}(X, Y) \leq B + \delta m \lfloor 4B/D_- \rfloor. \tag{B.11}$$

This proves membership in an upper norm window whenever the right side is below its cutoff; any lower endpoint condition is checked in the same way using (B.8). A product is subtracted only when this inclusion holds at every discriminant in the interval.

The factor lists are complete by the argument of Lemma B.2, applied at each admissible integer D in the interval. To remove a reducible candidate of norm at most F, one factor has norm at most $\sqrt{F}$ and the other at most F/m_D. Their lower endpoint norms are no larger, so the finite endpoint lists contain them. Equation (B.1) detects their product at that D. The initial factors below m_-^2 are covered by Lemma B.4. On the broad intervals used above 875, $P < m_-^3$, so only two-factor products are needed. On $[499, 503]$ the retained split-prime cutoff is below m_-^4, so two- and three-factor products suffice. Corrections to the row loss use only products proved to stay below P throughout the interval.

B.5. **The derivative remainder.** Legendre's formula applied to (B.7) gives

$$F(j) = \log(j!) - b(j) = \sum_{q} \sum_{a \geq 2} \lfloor j/q^a \rfloor \log q + \sum_{q \in \mathcal{Q}} \lfloor j/q \rfloor \log q. \tag{B.12}$$

Both sums are nondecreasing in j. Choose a nonnegative lower bound σ for

$$\sigma_{\mathrm{pp}} + \sum_{q \in \mathcal{Q}} \frac{\log q}{q}, \qquad \sigma_{\mathrm{pp}} = \sum_{\substack{q,\ a \geq 2 \\ q^a \leq J-1}} \frac{\log q}{q^a}.$$

The remaining task is to bound $H(j) = j\sigma - F(j)$ from above for every order $0 \leq j < J$.

Recall the set $\mathcal{S}$ defining $\mathcal{Q}$. Let $q_{\mathrm{cap}} \leq J - 1$ bound the upper endpoints of its norm intervals; for a fixed discriminant these intervals may be singletons. Let $\mathcal{C}_{\mathrm{out}}$ list every possible composite representative in $\mathcal{S}$, and let $\mathcal{C}_{\mathrm{in}}$ list representatives known to be composite and in $\mathcal{S}$ throughout the interval. Each product has multiplicity μ and a norm interval $[A, B]$. The preceding factor and inclusion arguments supply these two lists. Possible extra products in $\mathcal{C}_{\mathrm{out}}$ only weaken the lower bounds below. Then

$$\sigma_- = \sigma_{\mathrm{pp}} + \sum_{[a,b] \in \mathcal{S}} \frac{\log a}{b} - \sum_{\mathcal{C}_{\mathrm{out}}} \mu \frac{\log B}{A}, \tag{B.13}$$

$$\sigma_+ = \sigma_{\mathrm{pp}} + \sum_{[a,b] \in \mathcal{S}} \frac{\log b}{a} - \sum_{\mathcal{C}_{\mathrm{in}}} \mu \frac{\log A}{B}, \tag{B.14}$$

$$T_-(j) = \sum_{[a,b] \in \mathcal{S}} \lfloor j/b \rfloor \log a - \sum_{\mathcal{C}_{\mathrm{out}}} \mu \lfloor j/A \rfloor \log B \tag{B.15}$$

bound the slope from below and above, and the split-prime part of (B.12) from below. These inequalities follow by subtracting the composite contributions from the sum over all representatives. The containing list is used for a lower bound, and the contained list for an upper bound.

We use $\sigma = \sigma_- \geq 0$. On a block of integer orders $u \leq j < v$, monotonicity of the actual function F gives

$$H(j) \leq (v-1)\sigma_- - \sum_{q} \sum_{a \geq 2} \lfloor u/q^a \rfloor \log q - T_-(u). \tag{B.16}$$

Assign this upper bound to each order in the block, and sort the resulting J values as $h_{(1)} \geq \cdots \geq h_{(J)}$. With $M = nt + r$ as before,

$$c = \frac{2}{M} \left(n \sum_{i=1}^{t} h_{(i)} + r h_{(t+1)} \right) \quad \Longrightarrow \quad 2 \sum_{C} H(j) \leq Mc. \tag{B.17}$$

When $r = 0$, the last summand is omitted. This bound fills the largest values first, with capacity n per order. Blocks of at most 32 orders are used in the finite comparisons.

For the unbounded range it is useful to retain only prime powers $q^a \leq Q$, where Q is fixed. Omitting the others discards a favorable factorial contribution. For $d = q^a \leq Q$, put

$$b_d = n\lceil J/d \rceil, \quad t_d = \min(d, \lfloor M/b_d \rfloor), \quad r_d = M - b_d t_d.$$

Each residue modulo d occurs at most b_d times among the columns. Selecting residues in decreasing order therefore gives

$$\frac{2}{M} \sum_{C} \sum_{\substack{q,a \geq 2 \\ q^a \leq Q}} \{j/q^a\} \log q \leq 2 \sum_{d = q^a \leq Q} \frac{b_d \left(t_d(d-1) - \binom{t_d}{2} \right) + r_d \max(0, d - 1 - t_d)}{Md} \log q. \tag{B.18}$$

The estimate treats each prime power separately, so does not require simultaneously realizing the largest residues for different moduli.

B.6. **An inert prime in an auxiliary interval.** The following count supplies the prime used in Section 6. It uses the same norm representation, but requires only an upper bound for the number of split primes, so composite values need not be removed.

Lemma B.5. *Let $0 < A < B$, $16 < D_- \le D \le D_+$, and suppose $D \notin (A, B]$. Under class number one, the number of non-inert primes in $(A, B]$ is at most*

$$\sum_{y=1}^{\lfloor\sqrt{4B/D_-}\rfloor} \left(1 + \sqrt{B - D_-y^2/4} - \sqrt{\max(0, A - D_+y^2/4)}\right)_+. \tag{B.19}$$

Proof. Such a prime is split, so has a representative $q = X^2 + Dy^2/4$ with $X \ge 0$ and $y \ge 1$. For fixed y, the possible $X = x + y/2$ lie in a translate of $\mathbb{Z}$, and in the interval with endpoints $\sqrt{\max(0, A - D_+y^2/4)}$ and $\sqrt{B - D_-y^2/4}$. There are at most one plus its length. Also $D_-y^2/4 \le B$. Summing over y proves the bound. □

Appendix C. Additional spectral estimates

The estimates in Section 4 reduce the size of a selected minor to three bounds for a node Gram matrix. We first prove the complementary singular-product bound and the Schur bound. We then give finite estimates for the row weights and the three node bounds. All parameters remain free, so the same estimates apply when the polynomial space or the number of evaluation points changes.

C.1. **The complementary singular product.** The forward estimate uses an affine map taking the disk into itself. For a finite polynomial space, an expanding affine map is also available. To estimate its largest singular values, we pass to the inverse map and estimate the complementary product.

Proof of Lemma 4.3. The matrix T_J is triangular, with diagonal entries $1, d, \dots, d^{J-1}$. Consequently

$$\det(T_J^* T_J) = d^{J(J-1)}.$$

The product of its largest m squared singular values is this determinant times the product of the largest $k = J - m$ squared singular values of T_J^{-1}. The inverse is composition by $(z - c_0)/d$.

Put $D_{s_0} = \operatorname{diag}(s_0^j)_{0 \le j < J}$, with G_0, s_0 as in the lemma. Then $T_J^{-1} D_{s_0}$ is composition by

$$\psi(z) = -\frac{c_0 r_0}{G_0} + \frac{r_0}{G_0} z.$$

This map is a strict disk contraction. Indeed, G_0 is the larger root of

$$X^2 - (1 + r_0^2)X + (1 - c_0^2)r_0^2 = 0.$$

The left side at $X = (1 + c_0)r_0$ equals $-(1 + c_0)r_0(1 - r_0)^2 < 0$, so $G_0 > (1 + c_0)r_0$. Changing the sign of the translation does not change singular values, by the unitary reflection $f(z) \mapsto f(-z)$. Lemma 4.2, applied to ψ, gives squared singular values $G_0^b r_0^{2l}$, $l \ge 0$, for the infinite composition operator. Compression to polynomials of degree less than J cannot increase them.

For matrices X, Y, the norm inequality $\| \bigwedge^k (XY) \| \le \| \bigwedge^k X \| \, \| \bigwedge^k Y \|$ bounds the product of the largest k singular values of a product. Apply it to

$$T_J^{-1} = (T_J^{-1} D_{s_0}) D_{s_0}^{-1}.$$

The first factor contributes at most $G_0^{bk} r_0^{k(k-1)}$ to the squared singular product. If $s_0 \le 1$, the largest k diagonal entries of $D_{s_0}^{-1}$ have indices $J - k, \dots, J - 1$; if $s_0 \ge 1$, they have indices $0, \dots, k - 1$. Their squared products are respectively $s_0^{-k(2J-k-1)}$ and $s_0^{-k(k-1)}$. Multiplying these bounds and taking logarithms gives (4.11). □

C.2. **The Schur estimate.** The Schur bound sums all principal minors of a given size. Its proof has two parts. The Meixner factorization first compares each minor with a gamma moment determinant. Schur polynomials describe how that determinant depends on the selected orders, and the Cauchy identity sums this dependence over all selections.

We use the following standard notation. A partition of length at most m is an integer sequence $\lambda_1 \ge \cdots \ge \lambda_m \ge 0$; write $|\lambda| = \sum_i \lambda_i$. Its Schur polynomial is

$$s_\lambda(x_1, \dots, x_m) = \frac{\det[x_j^{\lambda_i+m-i}]_{i,j=1}^m}{\det[x_j^{m-i}]_{i,j=1}^m}.$$

The quotient extends to a symmetric polynomial when some variables coincide. At $1^m = (1, \dots, 1)$ it has value

$$s_\lambda(1^m) = \prod_{1\le i<j\le m} \frac{\lambda_i - \lambda_j + j - i}{j - i}. \tag{C.1}$$

We will use the Littlewood–Richardson product rule and the Cauchy identity in their usual Schur specializations; see [5] and [4, Section 2].

Lemma C.1 (Gamma moments)**.** *Let $b > 0$ and $m \ge 1$, let $0 \le j_0 < \cdots < j_{m-1}$, and put $D_T = \det[(b)_{j_i+j_k}]_{i,k=0}^{m-1}$. Then*

$$D_T \le \left(\prod_{0\le i<k<m} \frac{j_k - j_i}{k - i} \right)^2 \prod_{i=0}^{m-1} i!(b)_{2j_i-i}. \tag{C.2}$$

Proof. Let $\mathrm{d}\mu_b(x) = x^{b-1}e^{-x}\mathrm{d}x/\Gamma(b)$ on $(0,\infty)$. Its rth moment is $(b)_r$. Expanding both determinants and integrating term by term gives Andreief's identity in the form

$$D_T = \frac{1}{m!} \int_{(0,\infty)^m} \left(\det[x_k^{j_i}]_{i,k=0}^{m-1}\right)^2 \prod_{k=0}^{m-1} \mathrm{d}\mu_b(x_k).$$

Set $\lambda_{m-i} = j_i - i$. The alternant formula shows that $\det[x_k^{j_i}]$ is the Vandermonde determinant times $s_\lambda(x)$, up to a sign. For the consecutive orders $0, \dots, m-1$, the moment determinant is

$$D_0 = \det[(b)_{i+k}]_{i,k=0}^{m-1} = \prod_{i=0}^{m-1} i!(b)_i.$$

To see the last equality, factor $(b)_i$ from row i. The remaining entry $(b+i)_k$ is monic of degree k in $b+i$, so its determinant is the Vandermonde product $\prod_{i=0}^{m-1} i!$.

Write $\Delta(x) = \prod_{i<k}(x_k - x_i)$. Normalize the density $\Delta(x)^2 \prod_k \mathrm{d}\mu_b(x_k)$ to have mass one, and denote expectation for it by $\mathbb{E}$. The preceding calculation gives $D_T = D_0 \mathbb{E}[s_\lambda(X)^2]$. For any partition ν of length at most m, a second use of Andreief gives

$$\mathbb{E}[s_\nu(X)] = s_\nu(1^m) \prod_{i=1}^{m} (b + m - i)_{\nu_i}. \tag{C.3}$$

To evaluate it, Andreief writes the expectation as

$$\mathbb{E}[s_\nu(X)] = \frac{\det[(b)_{\nu_i+m-i+m-j}]_{i,j=1}^m}{D_0}.$$

Factor $(b)_{\nu_i+m-i}$ from row i. The remaining entries are the monic polynomials $(b+\nu_i+m-i)_{m-j}$. Dividing their Vandermonde determinant by the one for $\nu = 0$ gives $s_\nu(1^m)$ by (C.1); the ratios of the row factors give the rising factorials in (C.3).

The Littlewood–Richardson rule writes $s_\lambda^2 = \sum_\nu c_{\lambda\lambda}^\nu s_\nu$, with nonnegative coefficients. Every occurring ν has size $2|\lambda|$ and is dominated by 2λ: for every r, $\sum_{i\le r} \nu_i \le 2\sum_{i\le r} \lambda_i$. Terms with

more than m parts vanish in m variables. The decreasing vector $(b+m-i+2\lambda_i)_{i=1}^m$ therefore majorizes $(b+m-i+\nu_i)_{i=1}^m$. Convexity of $\log\Gamma$ yields

$$\prod_{i=1}^m (b+m-i)_{\nu_i} \le \prod_{i=1}^m (b+m-i)_{2\lambda_i}.$$

Apply (C.3) to the product expansion and evaluate that expansion at 1^m. We obtain

$$D_T \le D_0 s_\lambda(1^m)^2 \prod_{i=1}^m (b+m-i)_{2\lambda_i}.$$

Substituting $\lambda_{m-i} = j_i - i$ and using (C.1) gives (C.2). □

Proof of (4.12). The case $m = 0$ is immediate, so assume $m \ge 1$. Use the notation $M_{jk}(b,w)$, $\mathcal{B}$, σ_0, c_* and z_* from Section 4. The Meixner identities (4.6)–(4.7) give

$$M_{jk}(b,w) = (1-w)^{-b-j-k}(b)_j(b)_k \sum_{l=0}^{\min(j,k)} \frac{(j)_{\underline{l}}(k)_{\underline{l}}}{l!(b)_l} w^{j+k-l}. \tag{C.4}$$

These are finite consequences of the generating identity (4.5); the standard Meixner formulas are also recorded in [3, Sections 18.19 and 18.23].

Fix $T = \{j_0 < \cdots < j_{m-1}\}$, write $s = \sum_{j\in T} j$, and put $V_{lj} = (j)_{\underline{l}}/\sqrt{l!(b)_l}$ for $0 \le l, j < J$. Cauchy–Binet applied to (C.4) gives

$$(1-w)^{bm+2s} \det M[T,T] = \prod_{j\in T}(b)_j^2 \sum_{\substack{L\subseteq\{0,\dots,J-1\}\\ |L|=m}} \det(V_{L,T})^2 w^{2s-\sum_{l\in L} l}. \tag{C.5}$$

A nonzero term matches each $l \in L$ to an order $j \ge l$. Thus $\sum_{l\in L} l \le s$. After taking out w^s, the remaining polynomial has nonnegative coefficients and is at most its value at $w = 1$. The finite Vandermonde identity is

$$(b)_j(b)_k \sum_l \frac{(j)_{\underline{l}}(k)_{\underline{l}}}{l!(b)_l} = (b)_{j+k}.$$

For example, it follows by comparing the coefficient of x^j in $(1-x)^{-b-k} = (1-x)^{-b}(1+x/(1-x))^k$. Consequently (C.5), with the column and mass factors restored, gives

$$\det G[T,T] \le \mathcal{B}^m (1-w)^{-bm} c_*^{2s} D_T. \tag{C.6}$$

Write $a_* = J - m$ and $s_* = ma_* + m(m-1)/2$, and set $d_i = a_* + i - j_i$ for $0 \le i < m$. The inequalities on the j_i imply

$$a_* \ge d_0 \ge \cdots \ge d_{m-1} \ge 0.$$

Thus d is a partition, $s = s_* - |d|$, and the dimension factor in (C.2) is $s_d(1^m)$. The ratio $(b)_{2j_i-i}/(b)_{2a_*+i}$ removes $2d_i$ factors, all at least b. Combining Lemma C.1 with (C.6) therefore gives

$$\frac{\det G[T,T]}{\mathcal{B}^m(1-w)^{-bm}c_*^{2s_*}\prod_{i=0}^{m-1} i!(b)_{2a_*+i}} \le s_d(1^m)^2 z_*^{|d|}.$$

The assumption $bc_* > 1$ says $0 < z_* < 1$. Sum over the order sets T and enlarge the sum to all partitions of length at most m. The Schur Cauchy identity, specialized to two lists of m variables all equal to $\sqrt{z_*}$, gives

$$\sum_d s_d(1^m)^2 z_*^{|d|} = (1-z_*)^{-m^2}.$$

This is precisely the remaining factor in (4.12). □

C.3. **The signed row sum.** We now estimate the exact formulas at finite R and S. Throughout, $R = KS$, $1 \le J \le R$, and the weights are (4.2), with $a > 0$, $0 < v < 1$, and $\beta < 0$. The estimates below keep the row sum separate, since it can be negative.

Write $L(x) = x \log x$, with $L(0) = 0$, and define

$$f(x) = L(x) - L(a+x) + L(a) - x \log v, \qquad C_v = -a \log(1-v).$$

Lemma C.2. *Suppose $aR > 1$ and $f(1) < 0$. Then*

$$\frac{1}{W} \sum_{\Lambda} \lambda_{r,s} \le 3 \int_0^1 (1-x) f(x) \, \mathrm{d}x + \frac{9C_v}{R} + \frac{\beta}{2K}(1 - S^{-2}) + \frac{3(1+S^{-1})}{2R} \delta_R, \tag{C.7}$$

where

$$\delta_R = \log(R+1) + 1 + \log \frac{a+1}{a - 1/R}.$$

Proof. Integral comparison gives, for $r \ge 0$ and $b > 1$,

$$\log(r!) \le r \log r - r + \log(r+1) + 1, \qquad \log((b)_r) \ge \int_0^r \log(b+t) \, \mathrm{d}t - \log \frac{b+r}{b-1}.$$

For the second inequality, compare the sum with the integral over $[-1, r-1]$ and bound the two endpoint intervals. Substituting $b = aR$ in the row weights yields $\lambda_{r,s} \le R f(r/R) + \beta s + \delta_R$.

The function f is convex, $f(0) = 0$, and its minimum on $[0, \infty)$ is $-C_v$. Thus $f \le 0$ on $[0,1]$ and its total variation there is at most $2C_v$. The function $(1-x)f(x)$ has total variation at most $3C_v$: apply the product bound for variation, using $|f| \le C_v$. There are $S - \lfloor r/K \rfloor \ge S(1 - r/R)$ rows with first exponent r. Since f is nonpositive, their contribution is at most $RS(1-r/R)f(r/R)$. The left Riemann sum differs from the integral by at most $3C_v/R$, giving the first two terms of (C.7) after division by $W = R^2 S/3$.

Finally, $\sum_\Lambda s = KS(S^2-1)/6$ and $|\Lambda| = R(S+1)/2$. Summing $\beta s + \delta_R$ gives the last two terms of (C.7). □

The integral in (C.7) is elementary; it involves only integrals of $x \log x$ and $x^2 \log x$, after translation by a.

C.4. **Column capacities and the forward estimate.** Put $\varepsilon = J/R$. For a fixed upper squared radius $0 < q \le 1$ and a column scale $\tau > 0$, define

$$g_\tau(x) = L(x) + L(a+x) - L(a) + 2x(\ell - \log \tau - 1), \qquad E_{\mathrm{col}} = \frac{\log(R+1) + 1}{R}.$$

The price (4.8) satisfies

$$\frac{\nu_j}{R} \le g_\tau(j/R) + E_{\mathrm{col}}. \tag{C.8}$$

Indeed, use the factorial upper bound above and $\log((aR)_j) \le \int_0^j \log(aR + x) \, \mathrm{d}x$. The terms containing $\log R$ cancel those in the column scale, and $j + 1 \le J \le R$ bounds the factorial remainder.

The sum of the largest m column prices can be bounded without deciding which orders occur. Partition $[0, \varepsilon]$ into N_z equal bins. For a bin $I = [u, v]$, give it the upper cost $\gamma_I = \max\{0, g_\tau(u), g_\tau(v)\}$ and capacity

$$c_z = \frac{\varepsilon}{N_z} + \frac{1}{R}.$$

The function g_τ is convex, so γ_I bounds it throughout I. Each bin contains at most Rc_z of the possible points j/R. Assign boundary points to just one neighboring bin, and define

$$C_{q,\tau}(t) = \max \left\{ \sum_I \gamma_I u_I : 0 \le u_I \le c_z, \quad \sum_I u_I = t \right\}.$$

This function is nondecreasing and piecewise linear. Filling bins in decreasing order of γ_I evaluates it. Every selection of m orders gives a feasible choice at $t = m/R$; hence

$$\frac{\nu_m^*(\tau)}{R^2} \le C_{q,\tau}(t) + tE_{\mathrm{col}}.$$

The replacement of negative bin costs by zero ensures monotonicity and only enlarges the upper bound.

Retain $\delta_q, v_q, w, \mathcal{B}$ from (4.4). For the forward choice $0 < \xi < 1$, take Γ, τ as in Section 4, and set

$$d_{q,\xi} = a \log \frac{\Gamma}{1-w} + \frac{\delta_q}{K} + \frac{\log R}{R} + E_{\mathrm{col}} - \frac{\log \xi}{R}.$$

Since $\log \mathcal{B}/R \le \log R/R + \delta_q/K$, the forward bound (4.10) gives

$$\frac{F_q^{\mathrm{f}}(m)}{R^2} \le C_{q,\tau}(t) + d_{q,\xi} t + t^2 \log \xi. \tag{C.9}$$

The last term in $d_{q,\xi}$ retains the difference between $m(m-1)$ and m^2.

To use (4.13), we need a bound throughout an interval of possible values of t, rather than only at its endpoints. On a cell $[l, h] \subseteq [0, \varepsilon]$, a bound for $F_q^{\mathrm{f}}(m)/R^2 - a_0 t$ is

$$C_{q,\tau}(h) + \max\{(d_{q,\xi} - a_0)l, (d_{q,\xi} - a_0)h\} + l^2 \log \xi. \tag{C.10}$$

The three terms follow respectively from the monotonicity of $C_{q,\tau}$, the sign of the linear coefficient, and $\log \xi < 0$.

C.5. **Finite form of the complementary estimate.** For $\kappa > 0$, put

$$d = \frac{\kappa}{1+\sqrt{w}}, \qquad c_0 = d\sqrt{w}, \qquad \tau = \frac{d(1-w)}{\sqrt{v_q}}.$$

These are compatible with the affine map (4.9). Choose $0 < r_0 < 1$ and use G_0, s_0 from Lemma 4.3. Define

$$\begin{aligned} c_1 &= -a \log G_0 - \varepsilon(2 \log r_0 - \log s_0 - |\log s_0|), \\ c_2 &= \log r_0 - |\log s_0| < 0, \\ e_q &= -a \log(1-w) + \frac{\delta_q}{K} + \frac{\log R}{R} + E_{\mathrm{col}} + \frac{\max(0, -\log d)}{R}. \end{aligned}$$

The complementary bound for the node minor, divided by R^2, is at most

$$C_{q,\tau}(t) + \varepsilon(a+\varepsilon) \log G_0 + (c_1 + e_q)t + c_2 t^2. \tag{C.11}$$

To verify the finite term, substitute $J = \varepsilon R$, $m = tR$ and $k = (\varepsilon - t)R$ into (4.11). Using $\log s_0 = \log d + \log r_0 - \log G_0$, both branches become

$$\varepsilon(a+\varepsilon) \log G_0 + c_1 t + c_2 t^2 - \frac{t \log d + (\varepsilon - t) \log G_0}{R}.$$

Here $G_0 \ge 1$ and $0 \le t \le \varepsilon$, so the last term is at most $t \max(0, -\log d)/R$. Restoring the mass and column prices gives (C.11).

On a cell $[l, h]$, subtract $a_0 t$ and use

$$\begin{aligned} C_{q,\tau}(h) + \varepsilon(a+\varepsilon) \log G_0 + \max\{(c_1 + e_q - a_0)l, (c_1 + e_q - a_0)h\} \\ + c_2 l^2. \end{aligned} \tag{C.12}$$

As in the forward estimate, every term bounds the whole cell.

C.6. **The finite error in the Schur estimate.** At a positive upper squared radius q, put

$$\zeta = \ell + \tfrac{1}{2}\log v_q - \log(1-w), \qquad z_* = a^{-2}e^{-2\zeta}, \qquad I(x) = \tfrac{1}{2}x^2\log x - \tfrac{1}{4}x^2,$$

with $I(0) = 0$. In (4.12), the parameters of the completed Gram matrix satisfy $c_* = e^{\zeta}/R$ and $bc_* = ae^{\zeta}$. Define

$$\begin{aligned} g_\varepsilon(t) = {} & t(\delta_q/K - a\log(1-w)) + (2\varepsilon t - t^2)(\zeta - 1) + I(t) \\ & + I(a+2\varepsilon - t) - I(a+2\varepsilon - 2t) - ta\log a - t^2\log(1-z_*). \end{aligned} \tag{C.13}$$

Proposition C.3. *Suppose $R > 3$, $0 < a \le 1$, $0 < \varepsilon = J/R \le 1$, $\varepsilon(a+2\varepsilon) \le 1$, and $ae^{\zeta} > 1$. For $0 \le m \le J$ and $t = m/R$, the logarithm of the right side of (4.12), divided by R^2, is at most*

$$g_\varepsilon(t) + E_0, \qquad E_0 = \frac{\varepsilon}{R}(2\log R + 2 - \log\varepsilon - \log a). \tag{C.14}$$

Proof. The assertion is immediate for $m = 0$. For $m \ge 1$, put $a_* = J - m$, as in (4.12). In the product there, use

$$\log(i!) \le i\log i - i + \log(i+1) + 1, \qquad \log((aR)_n) \le \int_0^n \log(aR + x)\,\mathrm{d}x.$$

For $i = 0, \dots, m-1$, set $x_i = i/R$ and take $n = 2a_* + i$. The terms containing $\log R$ have total coefficient $\sum_i(2a_* + 2i) = 2s_*$, and cancel those in $c_*^{2s_*} = \exp(2s_*(\zeta - \log R))$. The remaining sum, divided by R^2, is the left Riemann sum $R^{-1}\sum_{i=0}^{m-1} f_t(x_i)$, where

$$\begin{aligned} f_t(x) = {} & x\log x + (a + 2\varepsilon - 2t + x)\log(a + 2\varepsilon - 2t + x) \\ & - a\log a - 2\varepsilon + 2t - 2x. \end{aligned}$$

The other finite terms have simple upper bounds. Since $a \le 1$ and $ae^{\zeta} > 1$, we have $\zeta > 0$. Thus replacing $2s_*/R^2 = 2\varepsilon t - t^2 - t/R$ by $2\varepsilon t - t^2$ drops the nonpositive term $-t\zeta/R$. The factor $\mathcal{B}^m$ contributes $t\delta_q/K + t\log S/R$, whose second term is at most $\varepsilon\log R/R$. Since $i + 1 \le J \le R$, the factorial remainders contribute at most $\varepsilon(\log R + 1)/R$.

It remains to estimate the Riemann sum. For $0 < x \le t$,

$$f_t'(x) = \log\bigl(x(a + 2\varepsilon - 2t + x)\bigr) \le 0,$$

because $x \le t \le \varepsilon$ and $\varepsilon(a + 2\varepsilon) \le 1$. Its total variation is at most

$$-\int_0^t (\log x + \log a)\,\mathrm{d}x = t(1 - \log t - \log a) \le \varepsilon(1 - \log\varepsilon - \log a).$$

The last inequality follows by differentiation, since $at \le 1$. For a decreasing function, the left Riemann sum exceeds the integral by at most its variation divided by R. Finally, direct integration gives

$$\begin{aligned} \int_0^t f_t(x)\,\mathrm{d}x = {} & I(t) + I(a + 2\varepsilon - t) - I(a + 2\varepsilon - 2t) \\ & - ta\log a - 2\varepsilon t + t^2. \end{aligned}$$

Combining this integral with the mass, $c_*^{2s_*}$, and $(1 - z_*)^{-m^2}$ gives (C.13). The three error bounds sum to E_0, proving (C.14). □

To obtain a bound on a whole cell, we use concavity. Differentiating (C.13) twice gives, for $t > 0$,

$$g_\varepsilon''(t) = \log\frac{t(a + 2\varepsilon - t)}{e^{2\zeta}(a + 2\varepsilon - 2t)^4(1 - z_*)^2}.$$

In particular, the sufficient condition

$$\varepsilon(a + \varepsilon)z_* < a^2(1 - z_*)^2 \tag{C.15}$$

makes g_ε concave on $[0,\varepsilon]$. Indeed, the numerator is at most $\varepsilon(a+\varepsilon)$, the fourth power in the denominator is at least a^4, and $e^{2\zeta} = 1/(a^2 z_*)$. For a cell with midpoint $t_0 > 0$ and half-width h_0, its tangent therefore gives the upper bound

$$g_\varepsilon(t_0) - a_0 t_0 + h_0 |g'_\varepsilon(t_0) - a_0| + E_0 \tag{C.16}$$

for $g_\varepsilon(t) - a_0 t + E_0$ throughout the cell. This also applies to the first cell, whose left endpoint is zero, by continuity.

C.7. **Combining the finite estimates.** Partition $[0,\varepsilon]$ into N_m cells. At each positive upper radius q, the forward and complementary formulas give bounds on every cell. Include (C.16) whenever the hypotheses of Proposition C.3 and (C.15) hold. For each cell, take the minimum over the available bounds and their parameter choices. Let $\Phi(q)$ be the maximum of these cell bounds and zero. Each chosen bound holds throughout its cell. Taking $F_q(m)$ to be the minimum of the corresponding exact node bounds, we obtain

$$F_q(m) \le a_0 R m + R^2 \Phi(q) \qquad (0 \le m \le J).$$

Thus Φ satisfies (4.13). Radial monotonicity (Lemma 4.1) and the node count in (4.15) now give the upper bound for the full minor. The signed row sum is supplied by (C.7).

For the finite comparisons, the additional parameter choices are

$$\kappa \in \{5/2, 3, 4, 6, 10\},$$

$$\xi, r_0 \in \{1/100, 3/100, 3/50, 1/10, 3/20, 1/5, 3/10, 2/5, \\ 1/2, 3/5, 7/10, 4/5, 9/10, 19/20, 99/100, 9999/10000\}.$$

The estimates themselves apply to any finite collections satisfying $\kappa > 0$ and $0 < \xi, r_0 < 1$. They may also use an upper bound for ℓ: each fixed principal minor is multiplied by $e^{2\ell \sum_{j\in T} j}$, so increasing ℓ enlarges it. All radial upper endpoints are positive; the original node minors at $q = 0$ are covered by Lemma 4.1.

Appendix D. Resultants and full rank

For the range $D \ge 875$ we use a rank criterion that does not require an auxiliary inert prime. The nodes lie on a small number of horizontal lattice lines. A polynomial of small weighted degree can vanish at few points of those lines unless it contains one of them as a factor; a resultant bounds the additional multiplicity at the remaining points.

For a polynomial $F(z,w)$, write

$$\deg_{S,R} F = \max\{Si + Rj : z^i w^j \text{ occurs in } F\}.$$

This degree is additive under multiplication, since the highest weighted homogeneous parts multiply to a nonzero polynomial. A polynomial supported on Λ has weighted degree at most $S(R-1) < RS$.

Lemma D.1. *Let $F \in \mathbb{C}[z,w]$ be irreducible, of z-degree $d \ge 1$ and weighted degree $H = \deg_{S,R} F$. Then $\mathcal{R}(w) = \operatorname{Res}_z(F, \partial_z F)$ is nonzero and*

$$\deg \mathcal{R} \le \frac{(2d-1)H - Sd^2}{R} \le \frac{H^2}{RS}.$$

If $F(z, w_0)$ has a root of multiplicity m at z_0, then $\mathcal{R}$ vanishes to order at least $m-1$ at w_0.

Proof. Write $F = \sum_{i=0}^{d} c_i(w) z^i$. Irreducibility and characteristic zero imply that F and $\partial_z F$ are relatively prime over $\mathbb{C}(w)$. Their resultant is therefore nonzero. Also $R \deg c_i \le H - Si$.

Each monomial in the Sylvester determinant has coefficient degree $2d-1$, and the sum of its coefficient indices is d^2. For example, the latter follows by replacing z by tz: the resultant of

$F(tz, w)$ and its z-derivative is $t^{d^2}\mathcal{R}(w)$. Thus each term has w-degree at most $((2d-1)H - Sd^2)/R$. The second inequality follows from

$$H^2 - S\big((2d-1)H - Sd^2\big) = (H - Sd)^2 + SH \geq 0.$$

It remains to justify the multiplicity statement, including when the leading coefficient vanishes at w_0. The specialization $F(z, w_0)$ is not identically zero: otherwise $w - w_0$ would divide the irreducible polynomial F, contrary to $d \geq 1$. Every row of its specialized Sylvester matrix is a polynomial divisible by $(z - z_0)^{m-1}$. The $m - 1$ functionals given by evaluation of the first $m - 1$ Taylor coefficients at z_0 therefore annihilate its row space. They are independent, so the matrix has corank at least $m - 1$. For a polynomial matrix, corank at least k at w_0 forces its determinant to be divisible by $(w - w_0)^k$: constant row and column operations put its value at w_0 in rank normal form, and each of the remaining rows supplies a factor $w - w_0$ in the determinant expansion. This proves the assertion without a condition on $c_d(w_0)$. □

Let the nodes lie in $|z| \leq \rho$. Write M_y for an upper bound on the number of horizontal lattice lines meeting this disk, and M_x for an upper bound on the number of nodes on any one such line. We may take

$$M_y = 2\lfloor 2\rho/\sqrt{D}\rfloor + 1, \qquad M_x = \lfloor 2\rho\rfloor + 1.$$

On a line of imaginary coordinate $y\sqrt{D}/2$ we have $w = \bar{z} = z - y\sqrt{-D}$.

Lemma D.2. *Let F be as in Lemma D.1. At a node z_i, let m_i be the multiplicity of z_i as a root of $F(z, \bar{z}_i)$. If F is a constant multiple of $z - w - y\sqrt{-D}$ for some integer y, then $H = R$ and $\sum_i m_i \leq M_x$. Otherwise*

$$\sum_i m_i \leq \frac{M_y H}{S} + \frac{H^2}{RS}.$$

Proof. A horizontal-line factor has simple z-roots and vanishes at nodes of one line only. In the other case, irreducibility implies that $F(z, z - y\sqrt{-D})$ is nonzero for every integer y. Its ordinary degree is at most H/S, since $S \leq R$. There are consequently at most $M_y H/S$ nodes at which F vanishes.

For the excess multiplicities, apply Lemma D.1 at $w = \bar{z}_i$. These values of w are distinct, because conjugation is injective. Hence $\sum_i (m_i - 1)_+ \leq \deg \mathcal{R} \leq H^2/(RS)$. Adding the two bounds proves the result. □

Proposition D.3 (Resultant rank criterion)**.** *The interpolation matrix V has full row rank if*

$$nJ > R(S + M_y), \qquad \max(J, M_x) \leq R + KM_y. \tag{D.1}$$

Proof. Otherwise there is a nonzero polynomial F supported on Λ such that $F(z, \bar{z}_i)$ vanishes to order at least J at z_i for every node. Factor it over $\mathbb{C}$ as

$$F(z, w) = F_0(w) \prod_\ell F_\ell(z, w),$$

listing repeated irreducible factors separately. Put $d_0 = \deg F_0$. Let k count the horizontal-line factors and let H_* be the sum of the weighted degrees of all other factors involving z. Additivity gives

$$R(d_0 + k) + H_* < RS.$$

A root of F_0 affects at most one node. Truncating each node's multiplicity at J, these roots contribute at most Jd_0. Lemma D.2 and $\sum H_\ell^2 \leq (\sum H_\ell)^2$ give

$$nJ \leq Jd_0 + M_x k + \frac{M_y H_*}{S} + \frac{H_*^2}{RS}.$$

Write $\beta = H_*/(RS)$, so $0 \le \beta < 1$ and $d_0 + k < (1-\beta)S$. The right side is strictly less than

$$(1-\beta)S\max(J, M_x) + \beta M_y R + \beta^2 RS \le (1-\beta)R(S+M_y) + \beta R(S+M_y).$$

This contradicts the first inequality in (D.1). □

Appendix E. A radial estimate for the larger discriminants

For the range $D \ge 875$ we use an entrywise estimate in place of the spectral estimate of Section 4. The two estimates start from the same normalized matrix. Here the additional saving comes from keeping track of which nodes and derivative orders can occur together in a determinant term. A node used many times must supply smaller derivative orders, while nodes near the origin give stronger radial decay. We express these restrictions by a convex function of the numbers of columns selected at the different radii.

Throughout this appendix assume $K \ge 2$ and $S \ge 1$. Put $R = KS$, $M = R(S+1)/2$, $W = R^2S/3$, and $\theta = M/(nJ) \le 1$. The derivative scale is denoted by ℓ, as in (2.15). We assume $\ell \ge 0$ and put $\varepsilon = J/R$. The parameters in Appendix F satisfy these assumptions.

E.1. **Counting the ordered exponents.** Put

$$u(G) = 1 - \sqrt{1-G}, \qquad v(G) = \begin{cases} \sqrt{G/K}, & 0 \le G \le 1/K, \\ 1 - \sqrt{(1-1/K)(1-G)}, & 1/K \le G \le 1. \end{cases}$$

The second function is the quantile of $r+s$, after scaling by R, on the continuous triangle $r + Ks \le R$. The following bounds account for the discrete rows of the interpolation matrix.

Lemma E.1. *List the values of r and $r+s$ on Λ in nondecreasing order as $r_{(k)}$ and $z_{(k)}$, respectively. For $G_k = k/M$,*

$$Ru(G_k) - 1 \le r_{(k)} \le (R+K)u(G_k), \qquad z_{(k)} \ge Rv(G_k) - 2.$$

Proof. For $0 \le a \le R$, the number of rows with $r \ge a$, with a an integer, is

$$T_r(a) = \sum_{b=1}^{S} (Kb - a)_+ = \frac{K}{2}\{y^2 + y + \alpha(1-\alpha)\}, \quad y = S - a/K, \quad \alpha = \{a/K\}.$$

This follows by summing the arithmetic progression of positive terms. It implies

$$M(1-a/R)^2 \le T_r(a) \le M(1-a/(R+K))^2.$$

For the first inequality, subtracting its right-hand quadratic from $y^2 + y$ leaves $y(1-y/S) \ge 0$. For the second, write $a/K = q + \alpha$. After multiplication by $S+1$, the difference of the two sides, apart from the positive factor $K/2$, is $q(S+1-q-2\alpha) + S\alpha^2 \ge 0$. The endpoints $a = 0, R$ satisfy the same inequalities. Applying the lower tail bound at $a = r_{(k)} + 1$ and the upper tail bound at $a = r_{(k)}$ proves the two assertions about $r_{(k)}$.

For the joint exponent, let $N(t)$ count rows with $r + s \le t$, for an integer $0 \le t < R$. Let F be the inverse of v, extended by $F(x) = 1$ for $x \ge 1$. We show that $N(t) \le MF((t+2)/R)$. If $t \le S - 2$, the whole triangle $r + s \le t$ is contained in Λ, so

$$N(t) = \frac{(t+1)(t+2)}{2} \le \frac{S+1}{2S}(t+2)^2 = MF((t+2)/R).$$

If $t \ge S - 1$, put $h = K - 1$ and $A = R - t - 1$. For $A \ge 1$ the complementary count is

$$M - N(t) = \sum_{s=0}^{S-1} (A - hs)_+ \ge \frac{A^2}{2h} + \frac{A}{2}.$$

Here $1 \le A \le hS$. On this interval

$$A^2 + hA \ge (1 + 1/S)(A-1)^2 :$$

the difference is a concave quadratic in A and is nonnegative at both endpoints. The last lower bound is therefore at least $M\{1-F((t+2)/R)\}$. When $A=0$ the desired assertion is immediate. Taking $t=z_{(k)}$ and using $k\le N(t)$ proves the remaining bound. □

E.2. **The falling factorial and the derivative orders.** For $0\le t\le x\le 1$, define

$$f(x,t)=x\log x-(x-t)\log(x-t)-t,\qquad g(x,t)=\max_{0\le a\le \min(x,t)}\{a\ell+f(x,a)\},$$

with $0\log 0=0$. We also use $g(x,t)$ for $t>x$ with this same definition. Its maximum occurs at $a=\max(0,\min(t,x-e^{-\ell}))$. For $0\le j\le r<R$, integral comparison gives

$$\log\frac{(r)_j}{R^j}\le Rf\left(\frac{r+1}{R},\frac{j}{R}\right). \tag{E.1}$$

Indeed, on each unit interval the decreasing function $\log((r+1-t)/R)$ is at least its value at the right endpoint. This also proves the inequality when $j=r$.

Fix a set C of M columns, and write c_i for the number selected at node i. Replace its selected orders by $J-c_i,\dots,J-1$, and list the resulting multiset increasingly as j_k^+. If the original orders are listed as $j_{(k)}$, then

$$j_{(k)}\le j_k^+\le t_k,\qquad t_k=J-\frac{M-k}{n}. \tag{E.2}$$

The first assertion follows by ordering the columns at each node; the second follows because at most $n(J-a)$ orders can be at least a.

Consider a nonzero term in the expansion of a selected determinant. Every assigned derivative satisfies $j\le r$. The mixed derivative $f_{xt}=1/(x-t)$ is positive. Exchanging two oppositely ordered pairs therefore increases the sum of f; the exchange preserves feasibility, since the larger derivative was already assigned to the smaller row degree. Successive exchanges pair the increasingly ordered row degrees and derivatives. Increasing the row bound and then using that g is nondecreasing in its second argument gives

$$\sum_C\left(j\ell+\log\frac{(r)_j}{R^j}\right)\le R\sum_{k=1}^M g(x_k,j_k^+/R),\qquad x_k=\min\left(1,(1+1/S)u(G_k)+1/R\right). \tag{E.3}$$

The exchange argument extends to the boundary $x=t$ by continuity.

Put $h(x,t)=t\ell-g(x,t)$. For $\ell\ge 0$, this function is nonnegative, increases with t and ℓ, and decreases with x. For example $g_t=(\ell+\log(x-t))_+$ before the maximum becomes constant, so $0\le g_t\le\ell$. The other assertions follow either by differentiation at the maximizing argument or directly from the definition. Thus an admissible normalized factorial saving is any number S_{der} satisfying

$$0\le S_{\mathrm{der}}\le\frac{3}{M}\sum_{k=1}^M h(x_k,t_k/R). \tag{E.4}$$

It can be bounded by grouping consecutive indices: on a group with $x_k\le b$ and $t_k/R\ge a$, use its relative size times $3h(b,a)$. This requires only a finite sum, rather than a list of the M rows.

E.3. **An additional saving from the number of columns at a node.** Suppose $\theta\varepsilon<1/2$. Choose integers $0<m_1<\cdots<m_L<M/n$, put $\tau_r=m_r/J$, and suppose

$$a_r=1-\sqrt{\tau_r/\theta}-(1-\tau_r)\varepsilon>0,\qquad \nu_r=\ell+\log a_r>0.$$

These ν_r decrease with r: the derivative of $1-\sqrt{\tau/\theta}-(1-\tau)\varepsilon$ is $\varepsilon-1/(2\sqrt{\theta\tau})<0$ for $\tau<\theta$. Set $\nu_{L+1}=0$ and $\lambda_r=\nu_r-\nu_{r+1}$. Smaller nonnegative increments may be used throughout.

If $t_k\ge J-m_r$, then $G_k\ge 1-\tau_r/\theta$. The function $u(G)-\varepsilon(1-\theta(1-G))$ is increasing, since its derivative is at least $1/2-\theta\varepsilon>0$. It follows that $x_k-t_k/R\ge a_r$. On the interval between j_k^+

and t_k, the derivative of $Rg(x_k, t/R)$ is therefore at least the staircase $\sum_r \lambda_r \mathbf{1}_{t \ge J - m_r}$. Integration gives an improvement to (E.3). The improvement can be summed explicitly, because

$$\sum_k (t_k - J + m)_+ = \frac{nm^2 + m}{2},$$
$$\sum_k (j_k^+ - J + m)_+ = \frac{nm(m-1)}{2} - \frac{1}{2}\sum_i (m - c_i)_+ (m - c_i - 1).$$

Subtracting these identities, and using $\sum_i (m - c_i)_+ \le nm$, gives a difference at least $\frac{1}{2}\sum_i (m - c_i)_+^2$. Consequently

$$\frac{2}{W}\sum_C \left(j\ell + \log \frac{(r)_j}{R^j} \right) \le \gamma\ell - S_{\mathrm{der}} - \frac{1}{W}\sum_{i,r} \lambda_r (m_r - c_i)_+^2, \quad \gamma = \frac{2}{W}\left(MJ(1 - \theta/2) + \theta J/2 \right). \tag{E.5}$$

In obtaining the factorial saving we used $2RM/W = 3(1 + 1/S) \ge 3$.

E.4. **Radial weights and the convex estimate.** Order the nodes by modulus. Suppose N divides n, and divide the nodes into N consecutive groups of n/N nodes. Choose numbers $L_0 \ge \cdots \ge L_{N-1} \ge L_N = 0$ such that

$$\log(\rho/|z_i|) \ge L_b \qquad (bn/N < i \le (b+1)n/N).$$

At a zero node the assertion has its limiting meaning. The disk estimate (2.9) permits the choice

$$L_b = \log \frac{1 + \eta}{\sqrt{\min(1, (b+1)/N + 1/n)} + \eta}, \qquad \eta = \frac{\sqrt{D+4}/4}{\sqrt{n\sqrt{D}/(2\pi)}}. \tag{E.6}$$

The extra $1/n$ weakens the bound and is convenient when a common bound is used for several values of n. Choose an index b_0 and a number $\overline{L} \ge L_{b_0}$. Write

$$w_b = \frac{N}{nJ} \sum_{i \text{ in group } b} c_i, \qquad G_b(w) = \frac{1}{N\theta}\sum_{a=0}^{b} w_a.$$

Thus $0 \le w_b \le 1$ and $\sum_b w_b = N\theta$. Define

$$\Phi(G) = 3\int_0^G \left(v(t) - \frac{J+2}{R} \right)_+ dt, \tag{E.7}$$

$$\Delta = \frac{2}{W}\left\{ \binom{J+2}{3} - \binom{(J-S)_+ + 2}{3} \right\}. \tag{E.8}$$

In particular, when $J \ge S$,

$$\Delta = 3\left(\frac{J+1}{R} \right)^2 - \frac{3}{K}\frac{J+1}{R} + \frac{1}{K^2} - \frac{1}{R^2}.$$

On the polytope $\mathcal{P}_\theta = \{ w \in [0,1]^N : \sum_b w_b = N\theta \}$, put

$$\mathcal{F}(w) = \sum_{b=0}^{N-1} (L_b - L_{b+1}) \Phi(G_b(w)) + \frac{3\varepsilon}{2N\theta} \sum_{b=0}^{N-1} \left\{ L_{\max(b,b_0)} w_b^2 + \sum_{r=1}^{L} \lambda_r (\tau_r - w_b)_+^2 \right\} - \overline{L}\Delta. \tag{E.9}$$

Proposition E.2 (Radial determinant estimate)**.** *With the preceding choices, let $F_* \le \min_{\mathcal{P}_\theta} \mathcal{F}$. Then every maximal minor satisfies*

$$\frac{1}{W}\log|\det B_C(\ell)|^2 \le \gamma\ell - S_{\mathrm{der}} - F_* + \frac{2M\log M}{W}. \tag{E.10}$$

Proof. Consider a nonzero determinant term, and put $z = r + s$ for its assigned row. Since $z \ge r \ge j$, for a radial weight $l \ge 0$ and $\widetilde{l} = \min(l, \overline{L})$ one has

$$(z-j)l \ge (z-J)_+ l + (J-j)\widetilde{l} - (J-z)_+\widetilde{l}.$$

Indeed, subtracting the first term leaves $(\min(z,J)-j)l$, whose coefficient is nonnegative. This observation allows us to retain the full radial weight on the first term while bounding the possible negative correction by $\overline{L}$.

Among all assignments of rows to columns, pairing the smallest values of $(r+s-J)_+$ with the largest radial weights minimizes their sum. For any integer $h \le M$, Lemma E.1 and the right-endpoint integral inequality for an increasing function give

$$\frac{2}{W}\sum_{k=1}^{h}(z_{(k)}-J)_+ \ge \frac{2RM}{3W}\Phi(h/M) \ge \Phi(h/M).$$

Summation by parts over the node groups therefore gives the first sum in (E.9). At node i, the distinct derivative orders satisfy $\sum(J-j) \ge c_i(c_i+1)/2 \ge c_i^2/2$. Also

$$\sum_{\Lambda}(J-r-s)_+ \le \sum_{s=0}^{S-1}\binom{(J-s)_+ + 1}{2} = \binom{J+2}{3} - \binom{(J-S)_+ + 2}{3}.$$

Thus the clipped radial terms contribute at least $W^{-1}\sum_i c_i^2 L_{\max(b(i),b_0)} - \overline{L}\Delta$.

Combine this with (E.5). Convexity on each node group gives

$$\sum_{i \text{ in group } b} c_i^2 \ge \frac{n}{N}J^2 w_b^2, \qquad \sum_{i \text{ in group } b}(m_r - c_i)_+^2 \ge \frac{n}{N}J^2(\tau_r - w_b)_+^2.$$

Since $nJ^2/W = 3\varepsilon(1+1/S)/(2\theta)$, dropping the factor $1+1/S$ proves that the total normalized saving is at least $S_{\mathrm{der}} + \mathcal{F}(w)$. The modulus of each nonzero determinant term is bounded accordingly. There are at most $M! \le M^M$ terms; taking the square of their sum proves (E.10). □

The function Φ is convex, because its derivative is increasing. Every coefficient of the convex terms in (E.9) is nonnegative. Hence $\mathcal{F}$ is convex. In particular, any $w^* \in \mathcal{P}_\theta$ gives the explicit lower bound

$$F_* = \mathcal{F}(w^*) - \nabla\mathcal{F}(w^*)\cdot w^* + \min_{w\in\mathcal{P}_\theta}\nabla\mathcal{F}(w^*)\cdot w. \tag{E.11}$$

The last minimum is found by filling the coordinates with the smallest coefficients first, subject to the upper bound one and the prescribed sum. This follows by transferring mass from a larger coefficient to a smaller one whenever the latter coordinate is not full. No assertion that w^* minimizes $\mathcal{F}$ is required. For numerical evaluation its gradient is obtained directly from (E.9); the positive-part squares are continuously differentiable, including at their endpoints.

Appendix F. The range $D \ge 875$

We now apply the radial estimate to the part of the proof preceding the finite comparisons of Section 6. The argument has two parts. On an unbounded interval the arithmetic estimate grows faster than the size estimate, with an explicit positive difference at the left endpoint. Below that endpoint we keep the integer parameters fixed on each of finitely many adjoining intervals.

F.1. **Parameters and geometry on the unbounded interval.** Set

$$D_0 = 5\,760\,000, \quad X = (D-3)/4, \quad X_0 = (D_0-3)/4, \quad K = 31, \quad c_R = 23/10, \quad e = 29/500.$$

For $D \geq D_0$ choose

$$S = \lfloor c_R X/K \rfloor, \quad R = KS, \quad J = \lfloor eR \rfloor,$$
$$T = 106\,838, \quad R_0 = KT = 3\,311\,978, \quad J_0 = 192\,094.$$

Thus $S \geq T$, $R \geq R_0$ and $J \geq J_0$. Write $J_* = J_0 + 1$, $M_0 = R_0(T+1)/2$, and put

$$c_- = c_R - K/X_0, \qquad e_- = e - 1/R_0,$$
$$r_0 = \frac{4\sqrt{1.02K/e_-}}{\sqrt{2\pi T\sqrt{D_0}}} + \frac{1+\sqrt{1+4/D_0}}{T}, \qquad \gamma_0 = 1 + r_0 + 800e/T.$$

These give $\gamma_0 < 1.002779879 < 1.02$. Choose n by rounding $(1+r_0)RS/J$ strictly upward to a multiple of 800. Then

$$\frac{RS}{J} < n \leq \gamma_0 \frac{RS}{J}. \tag{F.1}$$

The disk bound gives $M_y \leq 4\rho/\sqrt{D}+1$, and substitution of $n \leq 1.02RS/J$ shows $M_y/S \leq r_0$. The selected n therefore satisfies $nJ > R(S+M_y)$. Indeed, the definition of n implies $n \leq \gamma_0 RS/J < 1.02RS/J$, which justifies the radius estimate used to bound M_y.

For later use put

$$\theta_- = 398891129/800000000, \qquad \theta_+ = (1+1/T)/2,$$
$$\nu_- = c_-/(Ke), \qquad \nu_+ = \gamma_0 c_R/(Ke_-),$$
$$\eta_+ = \frac{\sqrt{D_0+4}/4}{\sqrt{\nu_- X_0\sqrt{D_0}/(2\pi)}},$$
$$C_- = \tfrac{1}{2}\log(\nu_-/\pi), \qquad C_+ = \tfrac{1}{2}\log\left(\nu_+\sqrt{1+3/(4X_0)}/\pi\right) + \log(1+\eta_+).$$

Equation (F.1) gives

$$\theta_- \leq \theta \leq \theta_+, \quad \nu_- X \leq n \leq \nu_+ X, \quad \eta \leq \eta_+, \quad \tfrac{3}{4}\log X + C_- \leq \log\rho \leq \tfrac{3}{4}\log X + C_+. \tag{F.2}$$

For the first inequality use $\theta_- < 1/(2\gamma_0)$. For the third, the square of the bound for η is a constant times $(D+4)/((D-3)\sqrt{D})$, which decreases for $D > 4$. The last inequality follows by writing $D = 4X+3$ in the disk formula. In particular $\eta_+ < 0.022619737$. The remaining resultant hypothesis holds as well: $J < R$, and

$$\frac{2\rho+1}{R} \leq \frac{2e^{C_+}}{c_-}X^{-1/4} + \frac{1}{c_- X} < 1.$$

The right side decreases and is less than one at X_0. Thus $\max(J, M_x) \leq R + KM_y$, and the resultant criterion supplies a nonzero maximal minor throughout this interval.

F.2. **A uniform arithmetic lower bound.** Only primes at most X are used in this part of the argument. They are all inert by Lemma 2.3. We give the lower bound explicitly, including its dependence on the anchor T, so that its validity as D increases is visible.

For an integer $q \geq 4$, put

$$E_T(q) = \left[\frac{1}{Kq} - \frac{3}{2R_0} + \frac{3}{2R_0 q} - \frac{q}{2\sqrt{3}KT^3}\right]_+.$$

The floor-sum estimate for $T(S,q)$ gives $2KT(S,q)/W \geq E_T(q)$ when $S \geq T$. One can obtain it by expanding the quadratic sum (A.1); the only signed periodic term is bounded using $|t(1-t)(1-2t)| \leq$

$1/(6\sqrt{3})$, $0 \le t \le 1$. For $4 \le q \le S$, the positive term $q/(2R_0^2)$ absorbs the Bernoulli remainder in the displayed expression; for $q > S$ the sum is zero. Dropping the remaining nonnegative terms gives the displayed expression. It increases when T increases, since its terms depending on T are $-3(1-1/q)/(2KT) - q/(2\sqrt{3}KT^3)$.

For a prime $p < K$, write $a_p = K - p$ and define

$$W_p^{(0)} = \frac{1-T^{-2}}{K} + \sum_{p^{2l} \le T} E_T(p^{2l}), \qquad C(v) = v^2(3-2v), \quad e_* = J_*/R_0.$$

For each $v \in \{0, 1/200, \dots, 1\}$, put

$$v_- = (v - 1/T)_+, \qquad q_v = \frac{a_p}{K} v \min(1, v + 1/T).$$

Define the corresponding row gains and column loss by

$$\begin{aligned}
A_p(v) &= \frac{a_p}{Kp^2}[C(v_-) - 3/T]_+ + \frac{a_p}{Kp^3}[1 - C(v) - 6/T]_+ \\
&\quad + \min\left(0, \frac{3a_p(K + p - 1 - 2(p^2-1))}{2K^2p^2T}\right) - \frac{3a_p p^2}{2K^2T^2}, \\
B_p(v) &= \left[\frac{a_p}{K^2}\left\{\left(\frac{1}{p} - \frac{1}{p^2-1}\right)v_-^3 - \frac{3}{2T}\right\}\right]_+, \\
C_p(v) &= 3(1+1/T)q_v(1 - q_v\theta_-/2)\frac{p-1}{p}e_* \\
&\quad + \mathbf{1}_{v>0}\frac{3(1+1/T)\lfloor p^2/4 \rfloor}{2p\theta_-(J_0/R_0)R_0^2}.
\end{aligned}$$

The resulting small-prime weight is

$$W_p = W_p^{(0)} + \max\left(0, \max_v\{[A_p(v) + B_p(v)]_+ - C_p(v)\}\right).$$

These are lower bounds for the small-prime row weights in (A.18). To see the direction of each estimate, take $L = \lfloor vS \rfloor$ there, so that $v - 1/S \le L/S \le v$. The sums of $t(S-t)$ in the two parts of the triangle, after normalization, have limiting values $C(v)$ and $1 - C(v)$; summing t and t^2 bounds their errors by $3/S$ on either part. The constant term of the strip estimate contributes at least its negative part. The incomplete-strip loss is at most $3a_p p^2/(2K^2S^2)$. For the extra odd-power sum, integral comparison of the floors gives the lower bound in B_p. Finally, the proportion of selected rows is at most q_v and $q(1 - q\theta/2)$ increases for $0 \le q \le 1$; the column estimate therefore gives C_p. Replacing S by T and θ by θ_- only weakens these bounds. For the final incomplete-column remainder, the denominator is bounded using $JR \ge J_0R_0$, which follows directly from $J \ge J_0$ and $R \ge R_0$.

Before replacing S by T, the corresponding lower gains increase with S, while the losses decrease: $(v - 1/S)_+$ increases, $(a_p/K)v\min(1, v + 1/S)$ decreases, and C is increasing on $[0, 1]$. Thus each fixed choice of v gives a uniform lower bound. Taking the maximum over the grid preserves this bound; any additional tower terms are nonnegative.

Write $L_0 = \log X_0$, $u_- = 1/c_R$, $u_+ = 1/c_-$, and

$$E_M = -\gamma_{\text{Euler}} - \sum_p \frac{\log p}{p(p-1)}.$$

The explicit estimates of Rosser and Schoenfeld [7, Theorems 4 and 6], followed by partial summation, give, for $x \ge X_0$,

$$\sum_{p\le x}\frac{\log p}{p} \ge \log x + E_M - \frac{1}{2\log x},$$

$$\vartheta(x) \le x\left(1+\frac{1}{2\log x}\right),$$

$$\sum_{K\le p\le x} p\log p \ge \frac{x^2}{2}\left(1-\frac{3}{2\log x}-\frac{1}{3\log^2 x}\right) - 160000. \tag{F.3}$$

For the last inequality, integrate the bound for ϑ from 563 and use

$$\frac{d}{dt}\left(\frac{t^2}{2\log t}+\frac{t^2}{3\log^2 t}\right) \ge \frac{t}{\log t} \qquad (t\ge 563).$$

The constant 160000 also covers the omitted initial interval.

The large-prime row weight has the lower bound

$$\frac{S_r(R,K,p)}{KW} \ge \frac{1}{p} - \frac{3}{2R} + \frac{3K}{2Rp} + \frac{p}{2R^2} - \frac{3K}{2R^2} - \frac{p^2}{12\sqrt{3}R^3}.$$

The two terms containing $3K$ have nonnegative sum when $p \le X < R$. Also $p^2 \le Xp$. Applying (F.3) therefore shows that the normalized arithmetic term is at least

$$A_0 + \log(X/X_0), \tag{F.4}$$

where

$$\begin{aligned} A_0 = {} & \sum_{p<K} W_p \log p + \sum_{\substack{p\ge K, l\ge 1\\ p^{2l}\le T}} E_T(p^{2l})\log p \\ & + L_0 + E_M - \frac{1}{2L_0} - \sum_{p<K}\frac{\log p}{p} - \frac{3}{2c_-}\left(1+\frac{1}{2L_0}\right) \\ & + \left(\frac{u_-^2}{2} - \frac{u_+^3}{12\sqrt{3}}\right)\left\{\frac{1}{2}\left(1-\frac{3}{2L_0}-\frac{1}{3L_0^2}\right) - \frac{160000}{X_0^2}\right\}. \end{aligned} \tag{F.5}$$

The coefficient on the final line is positive. As X increases, its braced lower bound increases; the negative logarithmic errors decrease in modulus. This proves the asserted uniformity in (F.4). Evaluation gives

$$A_0 > 10.8131594547983. \tag{F.6}$$

F.3. **Uniform bounds for the derivative and radial terms.** Retain only prime powers $q = p^a \le 1000$, $a \ge 2$, and put

$$\sigma = \sum_{p^a\le 1000,\ a\ge 2}\frac{\log p}{p^a}, \quad \ell_- = \tfrac{1}{4}\log X_0 + \log c_- - C_+ - \sigma, \quad \ell_+ = \tfrac{1}{4}\log X_0 + \log c_R - C_- - \sigma.$$

By (F.2), the actual slope is between $\ell_- + \frac{1}{4}\log(X/X_0)$ and $\ell_+ + \frac{1}{4}\log(X/X_0)$. Discarding further factorial savings gives a valid upper bound for the determinant.

Each residue modulo q can occur in at most $n\lceil J/q\rceil$ selected columns. Dividing by M bounds this capacity by $a_q = (1+q/J_0)/(\theta_- q)$. Put $b_q = \min(q, \lfloor 1/a_q\rfloor)$ and $r_q = (1-b_q a_q)_+$. Filling the

largest remainders first proves that

$$c_j = 2 \sum_{q=p^a\le 1000,\ a\ge 2} \frac{\log p}{q}\left\{a_q\left(b_q(q-1)-\frac{b_q(b_q-1)}{2}\right)+r_q(q-1-b_q)_+\right\} < 60.016864219225 \tag{F.7}$$

is an admissible bound for the column remainder.

For the factorial saving, use 3200 intervals $[u,v]$ of equal length in $[0,1]$. On each interval take

$$a=(1-\theta_+(1-u)^2)e_-,\quad b=\min(1,(1+1/T)v+1/R_0),\quad \alpha=[(1-u)^2-(1-v)^2-1/M_0]_+.$$

The number of quantiles in this interval is at least αM. The monotonicities following (E.3), with $\ell=\ell_-$, therefore give

$$S_{\mathrm{der}} = 3\sum_{[u,v]} \alpha h(b,a) > 0.1803971924811329636. \tag{F.8}$$

All the inequalities used here improve with R,S,J at their stated lower bounds; no new derivative orders or prime powers are assumed to have the same residues as those at the endpoint.

We next put the convex estimate on a fixed domain. Set $N=800$, $b_0=480$, and $n_0=1\,842\,400$, which is a lower bound for n by (F.1). In (E.6) use η_+ and n_0, and denote the resulting numbers by L_b^0. Take

$$\overline{L}_0=\tfrac12\log(N/(b_0+1)),\qquad u_0=(J_*+2)/R_0,$$

and let Δ_0 be (E.8) evaluated at (R_0,T,J_*). For every $D\ge D_0$ the actual threshold $(J+2)/R$ is at most u_0. Also $\Delta\le\Delta_0$: substitute $(J+1)/R\le e+1/R_0$ in its quadratic expression, whose derivative is positive above $1/(2K)$, and compare with J_*/R_0. The resulting strict bound absorbs the omitted negative term $-1/R^2$.

Choose 120 integers

$$m_r=\lfloor .02J_0\rfloor+\left\lfloor\frac{r(\lfloor .46J_0\rfloor-\lfloor .02J_0\rfloor)}{119}\right\rfloor,\quad 0\le r<120,$$

and put $\tau_r=m_r/J_0$. Retain the initial segment on which

$$G_r=1-\frac{\tau_r+1/J_0}{\theta_-}>0,\quad a_r=1-\sqrt{1-G_r}-(1-\tau_r)J_*/R_0>0,\quad \nu_r=\ell_-+\log a_r>0.$$

Use successive differences of these decreasing ν_r, with the next value after the last retained cut equal to zero, and reduce the cuts to $\widetilde{\tau}_r=(\theta_-/\theta_+)\tau_r$. These choices give lower staircase levels for the cuts $\lceil\tau_r J(D)\rceil$ at every $D\ge D_0$; the extra $1/J_0$ accounts for rounding the cuts upward.

In (E.9) replace θ by θ_-, the radial coefficients by L_b^0, the threshold in Φ by u_0, and the quadratic coefficient $3\varepsilon/(2\theta)$ by $3(J_0-1)/(2R_0\theta_-)$. Use the cuts $\widetilde{\tau}_r$ and the preceding staircase levels, and replace the negative correction by $-\overline{L}_0\Delta_0$. Call the resulting function $\mathcal{F}_0$. For any actual occupancy vector w, put $\widetilde{w}=(\theta_-/\theta)w$. Then $\widetilde{w}\in\mathcal{P}_{\theta_-}$ and its cumulative fractions G_b are unchanged. Moreover,

$$\frac{\varepsilon}{\theta}w_b^2=\frac{\varepsilon\theta}{\theta_-^2}\widetilde{w}_b^2\ge\frac{J_0-1}{R_0\theta_-}\widetilde{w}_b^2.$$

The same calculation applies to the positive-part squares, since their rescaled cuts are at least $\widetilde{\tau}_r$. The radial weights and their successive differences decrease when either η or the shift $1/n$ increases; this follows by differentiating the logarithm in (E.6). Finally $\overline{L}_0$ is an upper bound for the clipping weight. Each of these replacements therefore weakens the lower bound, and

$$\mathcal{F}_D(w)\ge\mathcal{F}_0(\widetilde{w}). \tag{F.9}$$

The supporting-plane formula (E.11), applied to $\mathcal{F}_0$, gives

$$\min_{\mathcal{P}_{\theta_-}}\mathcal{F}_0 > 0.0825081841978627281 =: F_*. \tag{F.10}$$

The supporting-plane evaluation uses only the explicit function above and the finite linear minimization in (E.11); the repository code records the parameter values and evaluates it with outward-rounded arithmetic.

F.4. **The resulting contradiction.** Put

$$A = 1 + \frac{1}{K} + \frac{3(K-1)}{2KT} + \frac{K-5}{2KT^2}, \qquad B = \frac{3(T+1)J_*}{R_0T}(1-\theta_-/2) + \frac{3\theta_+J_*}{R_0^2T}.$$

These bound $(W_1+W_2)/W$ and γ in (E.5) from above. Combining the radial determinant bound with the arithmetic column remainder gives the Archimedean upper bound

$$U_0 + \tfrac{3}{4}(A+B)\log(X/X_0), \tag{F.11}$$

where

$$\begin{aligned} U_0 = A(\tfrac{3}{4}\log X_0 + C_+) + B\ell_+ - F_* - S_{\mathrm{der}} \\ + \frac{3(T+1)c_j}{2R_0T} + \frac{3(T+1)\log M_0}{R_0T} < 10.8128420196003. \end{aligned} \tag{F.12}$$

The coefficient of c_j decreases with S. So does the last term: $((S+1)/S^2)\log(KS(S+1)/2)$ has negative derivative for $S \geq T$, as follows by logarithmic differentiation and $\log(KS(S+1)/2) > 2$. The geometry grows by at most $3A\log(X/X_0)/4$, while the derivative slope grows by at most $B\log(X/X_0)/4$. The larger coefficient in (F.11) is thus valid.

Subtracting (F.11) from (F.4) yields

$$G_{\mathrm{tail}}(D) > 0.00031743 + 0.127829\log(X/X_0) > 0. \tag{F.13}$$

Here $A_0 - U_0 > 0.0003174351980$ and $1 - 3(A+B)/4 > 0.1278297612987827$. The nonzero minor supplied above must satisfy the opposite determinant inequality. This excludes every $D \geq D_0$.

F.5. **The finite interval below D_0.** For $875 \leq D \leq D_0$, keep P, K, S, R, J, n fixed on each closed interval $[d_-, d_+]$. The lattice radius is enclosed using its endpoint values. Choose n to be a multiple of 800 satisfying the resultant conditions with the upper radius and the lower value of D. The arithmetic estimates use norm representations when $P > (d_- - 3)/4$; otherwise every retained prime is inert.

In all cases use $N = 800$ radial groups, 3200 groups for the factorial saving, and 120 cuts in (E.5). The latter run from $\max(1, \lfloor .02J \rfloor)$ to

$$\max\left(\max(1, \lfloor .02J \rfloor) + 1, \min(\lfloor M/n \rfloor - 1, \lfloor .495J \rfloor)\right),$$

equally spaced with integer rounding, retaining the initial segment where the stated positivity conditions hold. Take $b_0 = 480$. The prime-power cutoff is $J - 1$.

The intervals using norm representations have the parameter families in Table 5; there $S = \lfloor c_R P/K \rfloor$, $R = KS$, $J = \lfloor eR \rfloor$. Their slope windows have cutoff at most 750000 in the first row and 140000 thereafter. The prime-product bounds in Appendix B apply on each constituent interval.

For the remaining intervals take $K = 31$, $c_R = 23/10$, $e = 29/500$, and $P = \lfloor (d_- - 3)/4 \rfloor$. Between 3968539 and 4894699 the prime sums are evaluated directly. Above 4894699 use the explicit bounds of [7, Theorems 4, 6 and 18]; the additional estimate $\vartheta(x) < x$ is used only for $x \leq 10^8$, within its stated range. On each interval the radial function is bounded below with the upper value of η, while the negative clipping term uses the lower value. The derivative scale likewise uses the appropriate endpoint in each term. These substitutions bound the entire interval.

The adjoining intervals cover $[875, D_0]$, and their lower bounds are summarized in Table 1. Each is positive. Together with (F.13), this excludes every $D \geq 875$. The interval endpoints and the numerical calculations are available with the accompanying code.

Range	P	K	c_R	e
$[875, 1167]$	8 999 999	23	7/5	7/125
$[1167, 1359]$	1 699 999	23	8/5	27/500
$[1359, 1871]$	1 699 999	23	9/5	27/500
$[1871, 6959]$	1 699 999	25	2	7/125
$[6959, 319591]$	1 699 999	29	11/5	29/500
$[319591, 3968539]$	1 699 999	31	23/10	29/500

TABLE 5. Parameter families for the norm-form comparisons. Each range is subdivided into closed intervals on which all integer parameters are fixed.

AI USE

The research for this paper was carried out with autonomous rStation agents using Gemini 4 Argon and GPT-6 Astra. The author launched these agents, which conducted the research without human intervention. The initial arguments were recorded as paper 2, version 12 (under the agent name Severi Jet), giving the bound 499, and paper 22, version 1 (under the agent name Mira Cairn), giving 498. The local arithmetic, resultant and global estimates used here are in Sections 3–6 of the former; the spectral estimates are developed in paper 16, version 2, Sections 2–5 and 6.4. The necessary derivations are included in the appendices. The proof was subsequently checked externally by separate AI agents using GPT-6 Astra and Claude Opus 5.5, run independently of the agents that made the discoveries.

GPT-6 Astra, Claude Opus 5.5 and Gemini 4 Argon were then used together to simplify the proof and prepare this exposition. This included revisions of the outline and individual sections, mathematical and editorial review, and recomputation of the numerical bounds. In particular, narrowing the auxiliary-prime interval gives the shorter rank argument proved in Section 5.

REFERENCES

[1] Alan Baker, *Linear forms in the logarithms of algebraic numbers*, Mathematika **13** (1966), no. 2, 204–216. doi:10.1112/S0025579300003971.

[2] Ben Green, *On Landau–Siegel zeros*, Hexagon preprint 2610.00311, version 2, 18 October 2026. https://hexagonmath.org/pdf/2610.00311v2.

[3] NIST Digital Library of Mathematical Functions, §§18.19 and 18.23, https://dlmf.nist.gov/18.19 and https://dlmf.nist.gov/18.23.

[4] P. J. Forrester and S. O. Warnaar, *The importance of the Selberg integral*, Bull. Amer. Math. Soc. (N.S.) **45** (2008), 489–534. doi:10.1090/S0273-0979-08-01221-4.

[5] William Fulton, *Young Tableaux*, London Mathematical Society Student Texts, vol. 35, Cambridge University Press, 1997. doi:10.1017/CBO9780511626241.

[6] Kurt Heegner, *Diophantische Analysis und Modulfunktionen*, Math. Z. **56** (1952), 227–253. doi:10.1007/BF01174749.

[7] J. Barkley Rosser and Lowell Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. **6** (1962), 64–94. doi:10.1215/ijm/1028998657.

[8] Harold M. Stark, *A complete determination of the complex quadratic fields of class-number one*, Michigan Math. J. **14** (1967), 1–27. doi:10.1307/mmj/1028999653.

[9] Terence Tao, *A sketch of the proof of non-existence of Siegel zeroes*, *What's New*, 19 October 2026.

[10] Terence Tao, *An experimental AI math challenge: using tenth discriminant bounds to explore arguments ruling out Siegel zeroes*, *What's New*, 20 October 2026.